Neoplasia

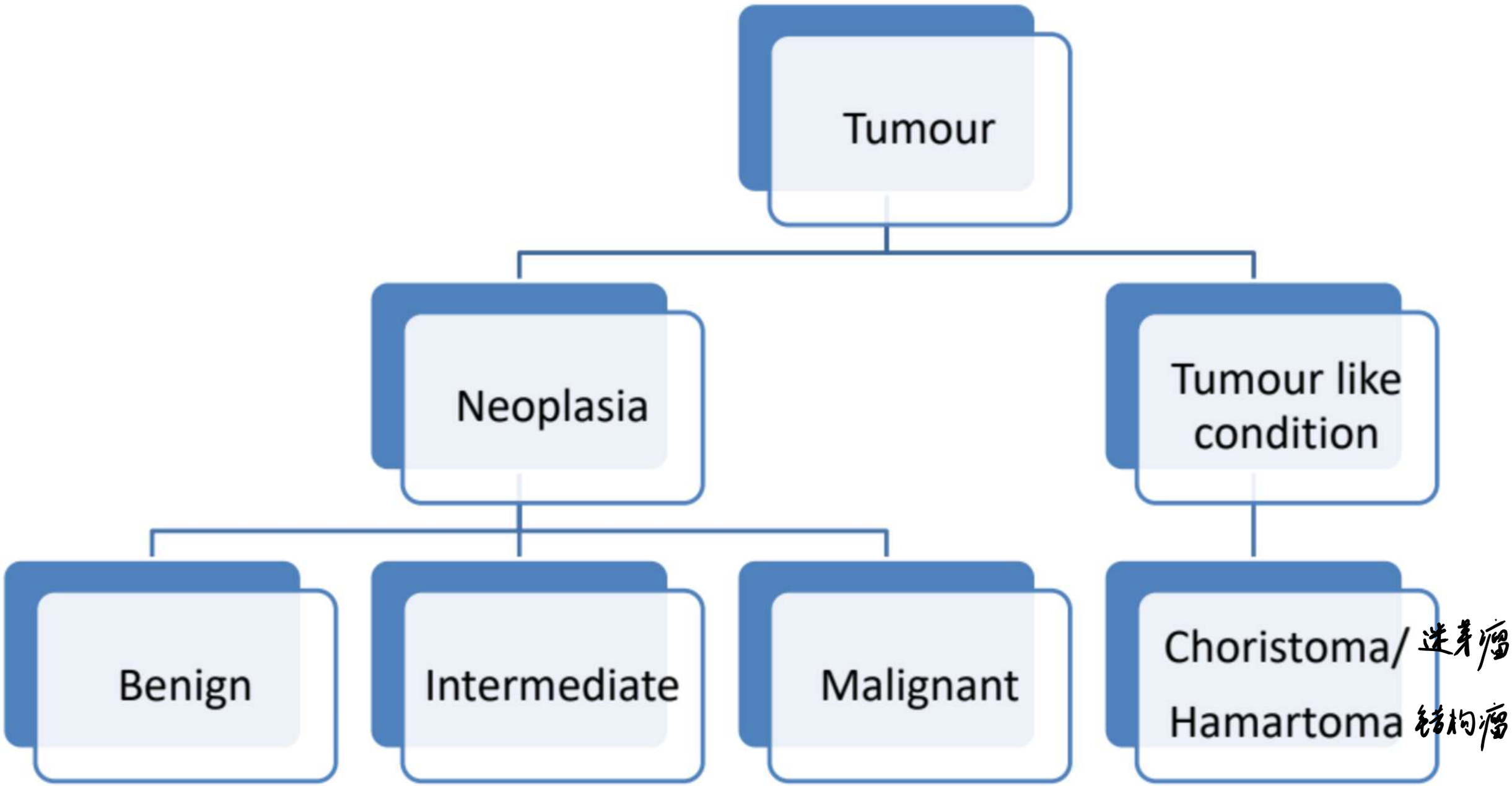
Dr Mohammad Zulkarnaen

- Part 1: terminology
- Part 2: Benign and malignant
- Part 3: Premalignant
- Part 4: Cancer incidence
- Part 5: Spread and effect
- Part 6: Prevention and tumour marker

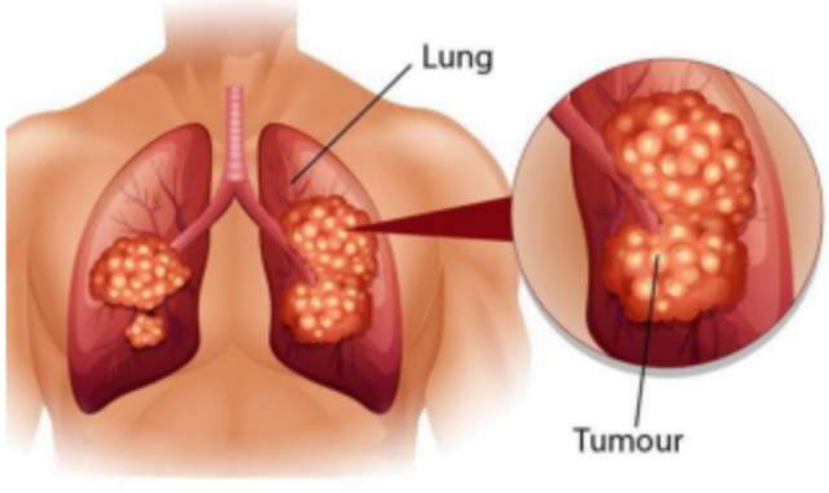
Part 1: Terminology

- Tumor is a general term used for describing Pathologic disturbance of growth, characterized by excessive and unnecessary proliferation of cells
- Galen (131 - 201 AD)
  - Swelling according to nature (pregnancy)
  - Swelling exceeding nature (Callus)
  - Swelling contrary to nature




Tumour
Neoplasia
Tumour like condition
Benign
Intermediate
Malignant
Choristoma/ 迷芽瘤
Hamartoma 错构瘤

- Neoplasia is “new growth”
- Neoplasm is defined as an abnormal mass of tissue characterized by growth that
  - Exceeds that of normal tissues
  - Is uncoordinated with that of normal tissues
  - Persists after cessation of inducing stimuli
  - **(R. A. Willis, D.Sc., M.D., F.R.C.P. Professor of Pathology, University of Leeds, 1953)**



INTRODUCTION

- Neoplasms can be divided into three categories based upon their behavior
  - Benign - neither invade, nor metastasize
  - Malignant - invade and often metastasize
  - Borderline - cannot predict behavior with certainty

Benign neoplasm

- Nomenclature
  - Suffix “–oma” is added to the cell type
  - From connective tissue (mesenchymal) origin arising from mesoderm
    - Cartilage: chondroma
    - Bone: osteoma
    - Adipose tissue: lipoma
    - Smooth muscle: leiomyoma
    - Skeletal muscle: rhabdomyoma
  - From epithelial origin arising from endoderm or ectoderm
    - Epithelial: adenoma (cystadenoma, papilloma)
  - If derived from more than one germ cell is called teratoma (contain tissue from mesoderm, endoderm and ectoderm)

- Ovarian cystadenoma
- pancreatic serous cystadenomas (honeycomb appearance) (never turn into cancer)
- Biliary cystadenomas (high risk turn into maglinancy)
- salivary glands

colon ~
- grow along colon polyps
- tubular, villous or tubulovillous

2. pituitary ~
- microadenoma (<1cm)
- macroadenoma (>1cm)

3. adrenal ~
- grow in adrenal gland
- overproduce hormone (cortisol/aldosterone)

4. parathyroid ~
- grow on parathyroid gland
- hyperparathyroidism

5. pleomorphic ~
- benign, affect salivary gland

6. hepatic ~
- rare benign tumor
- often associated with oral contraceptive use

subtype of
surface epithelium
serous
mucinous

- Both originated in glandular tissue, but cystadenoma is fluid-filled subtype of adenoma
- adenoma → solid mass/nodule (Eg. polyp/lump)
- cystadenoma → hollow, fluid-filled sac (cyst) lined with glandular cells

- arise from pluripotent germ cells (able to differentiate into any cell types)

Types:
1. mature teratoma (dermoid cysts)
- benign
- Ex. dermoid cyst of ovary

2. immature teratoma
- maglinant

3. monodermal/specialized teratoma
- composed of one specific tissue type
- Eg. struma ovarii (an ovarian teratoma made predominantly of thyroid tissue → produce thyroid hormone)
(can cause hyperthyroidism)
(5% maglinant)

Malignant tumors

- Nomenclature
  - Mesenchymal: **sarcoma** 肉瘤
    - Cartilage: chondrosarcoma
    - Bone: osteosarcoma
    - Adipose: liposarcoma
    - Smooth muscle: leiomyosarcoma
    - Skeletal muscle: rhabdomyosarcoma
  - Epithelial: **carcinoma**
    - Gland forming: adenocarcinoma
    - Squamous: Squamous cell carcinoma
    - Transitional cell: Transitional cell carcinoma (urothelial carcinoma)

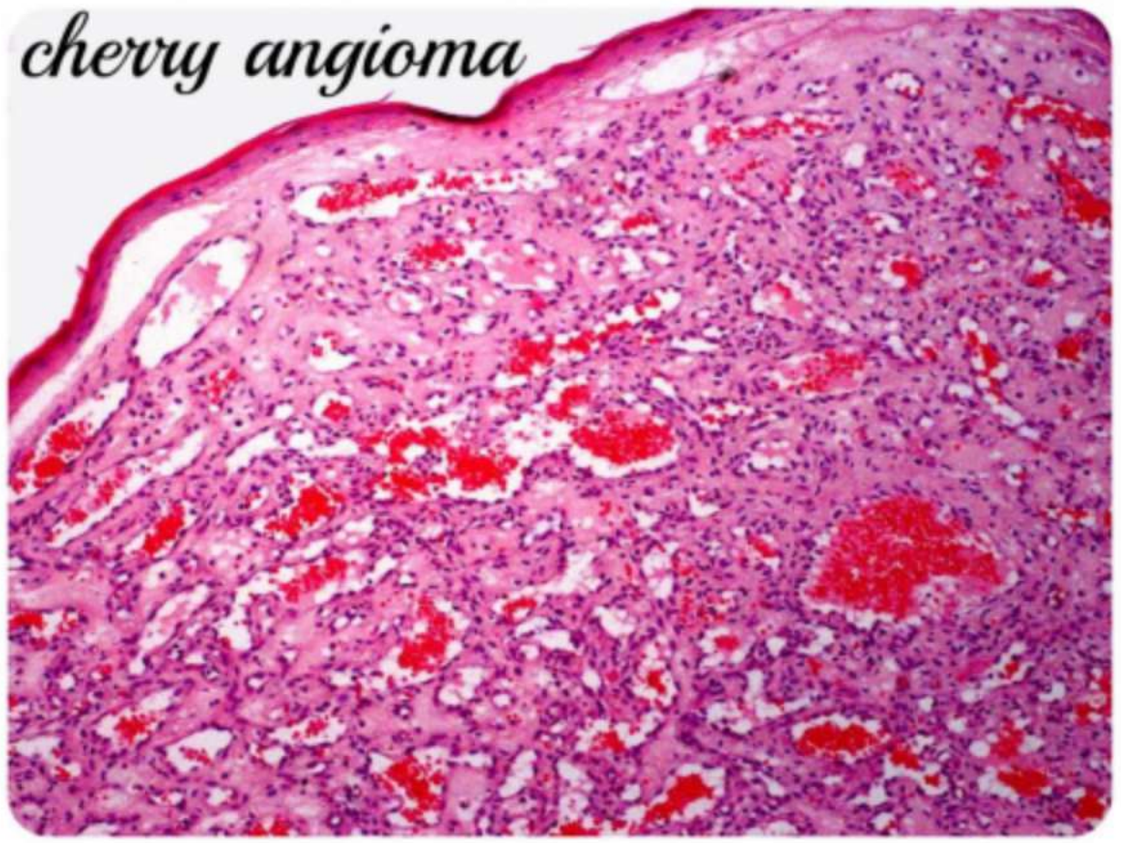
Tumour like conditions

- **Hamartoma** : Excessive but focal overgrowth of disorganized cells and tissues native to the organ in which it occurs.
  - Hamartoma of lung
  - Angiomas
  - Pigmented nevi
- **Choristoma (Heterotopia)** : Normal cells or tissues, that are present in abnormal locations
  - Pancreatic cells in the wall of stomach or intestine
  - Nests of adrenal cells in kidney , lung or ovaries

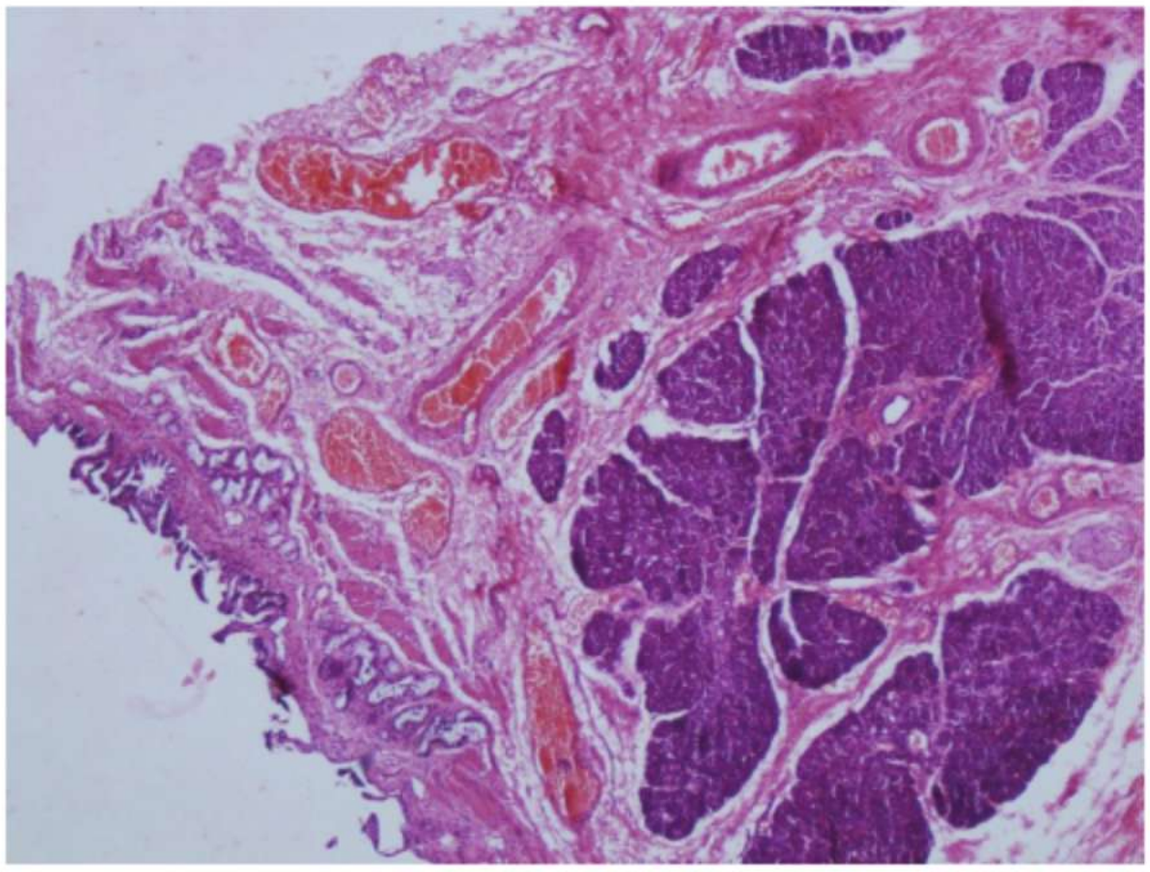
Example of hamartoma

Hamartoma : Excessive but focal overgrowth of disorganized cells and tissues native to the organ in which it occurs.





Example of choristoma



- Pancreatic cells in the wall of gallbladder

Choristoma (Heterotopia) : Normal cells or tissues, that are present in abnormal locations

Part 2: Benign and Malignant

- What are the characteristics of benign and malignant neoplasm?
- Features:
  - Differentiation and Anaplasia
  - Nuclear features
  - Rate of Growth
  - Local Invasion
  - Metastasis
  - Telomerase activity
  - Monoclonality

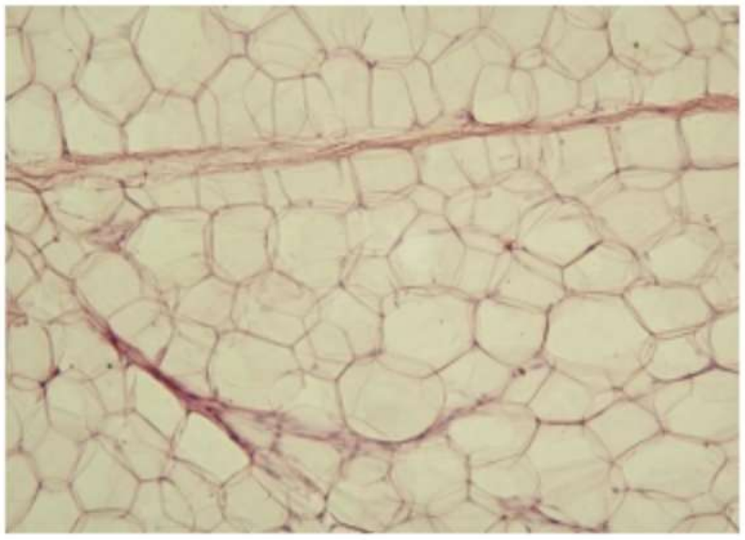
lack of normal cellular differentiation

Differentiation and Anaplasia

- Differentiation is the extent to which neoplastic cells resemble their normal cells morphologically and functionally
- Generally:
  - Benign neoplasm are well differentiated
  - Malignant neoplasm can be well differentiated, moderately differentiated or poorly differentiated. They can be “undifferentiated”
- Anaplasia is lack of differentiation

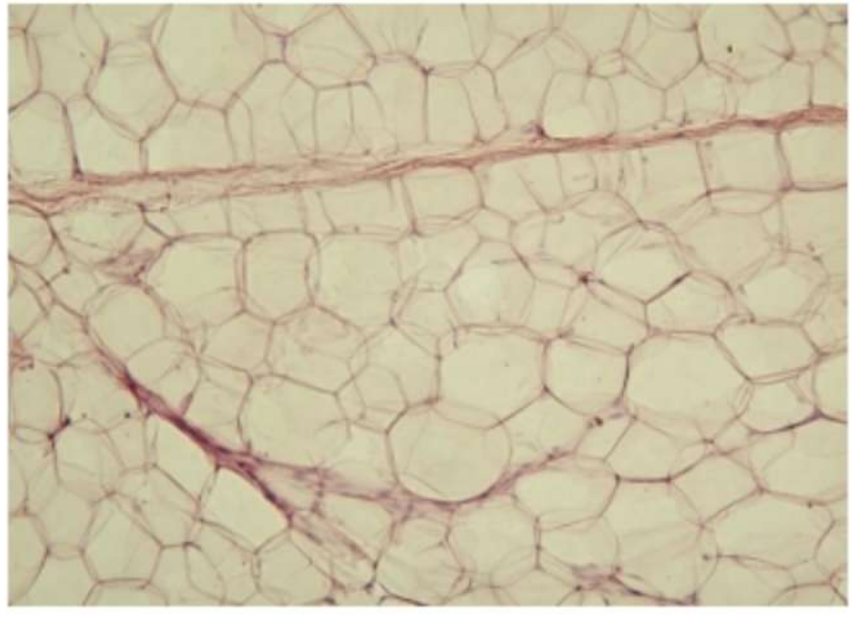
Well differentiated

Normal adipocytes



Lipoma



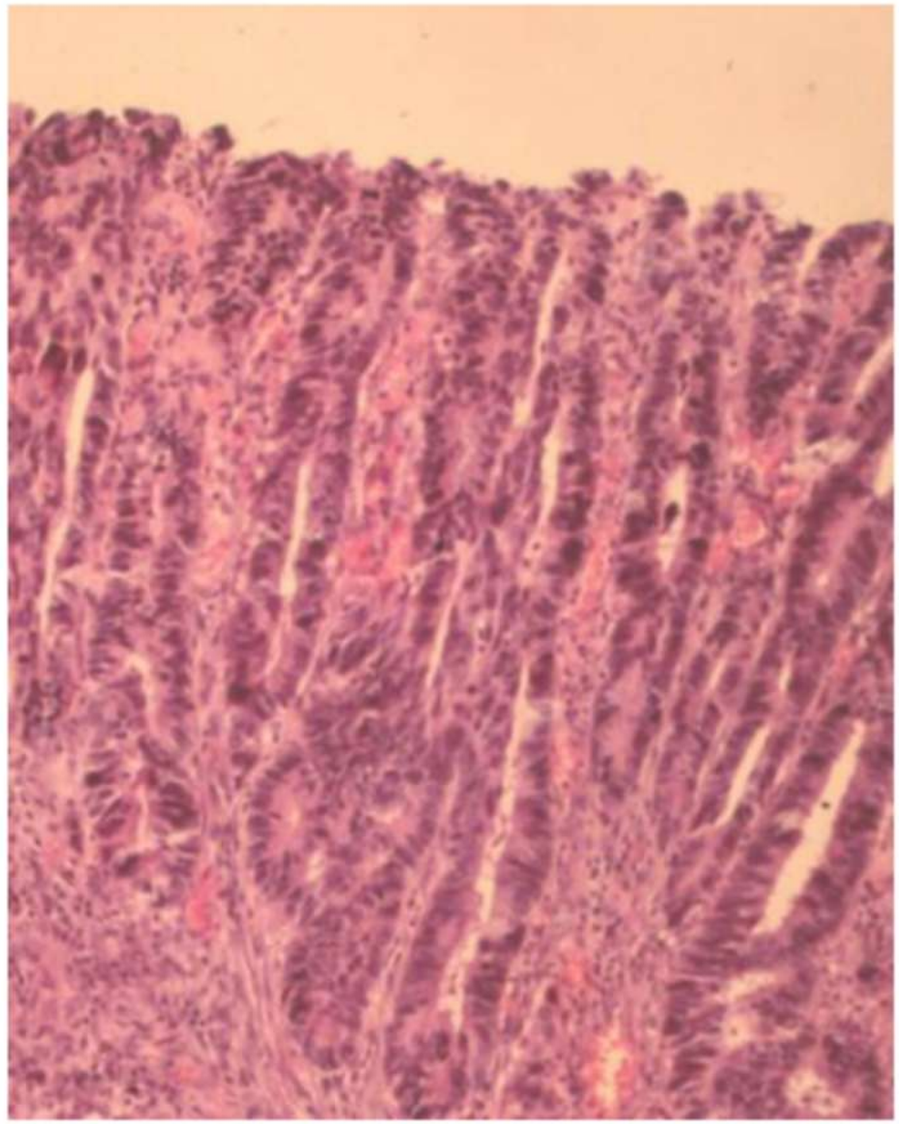


Now we look at the concept of well differentiated neoplasm. First we look at the normal adipocyte. Now look at the lipoma, which is a benign neoplasm. We mentioned generally benign neoplasm are well differentiated which means the neoplastic cells resemble their normal cells morphologically and functionally. Under the microscope, the lipoma cells will look very similar to the normal adipocyte cells.

Normal colonic mucosa

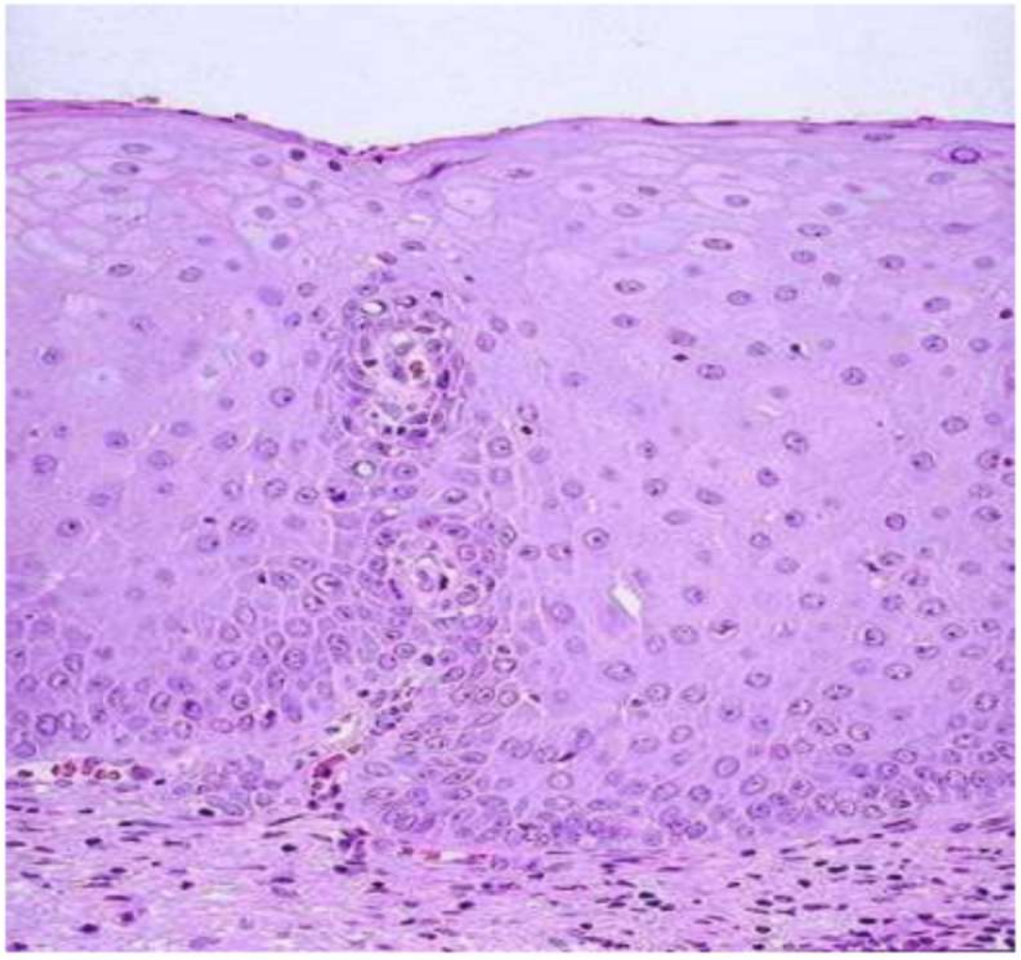


Neoplasia: Well differentiated adenocarcinoma

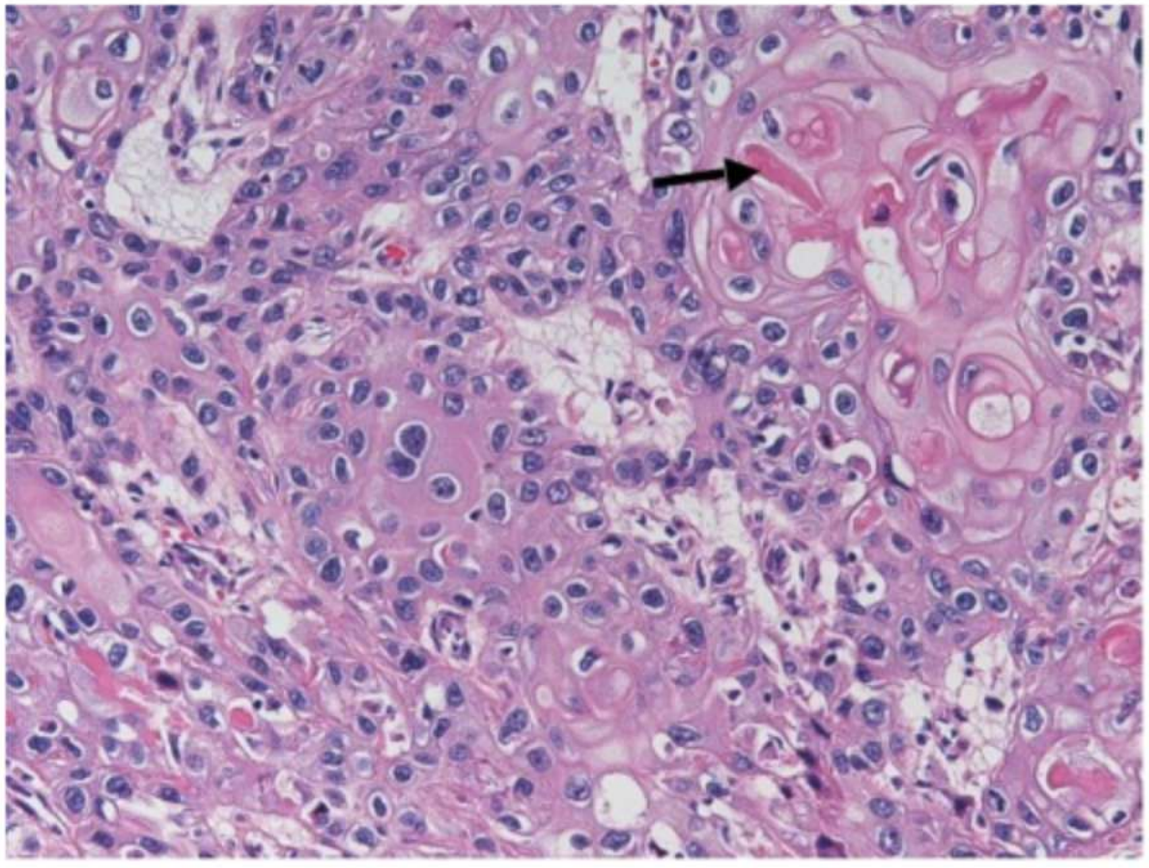


The neoplasia still forms identifiable glandular structure with presence of lumen,

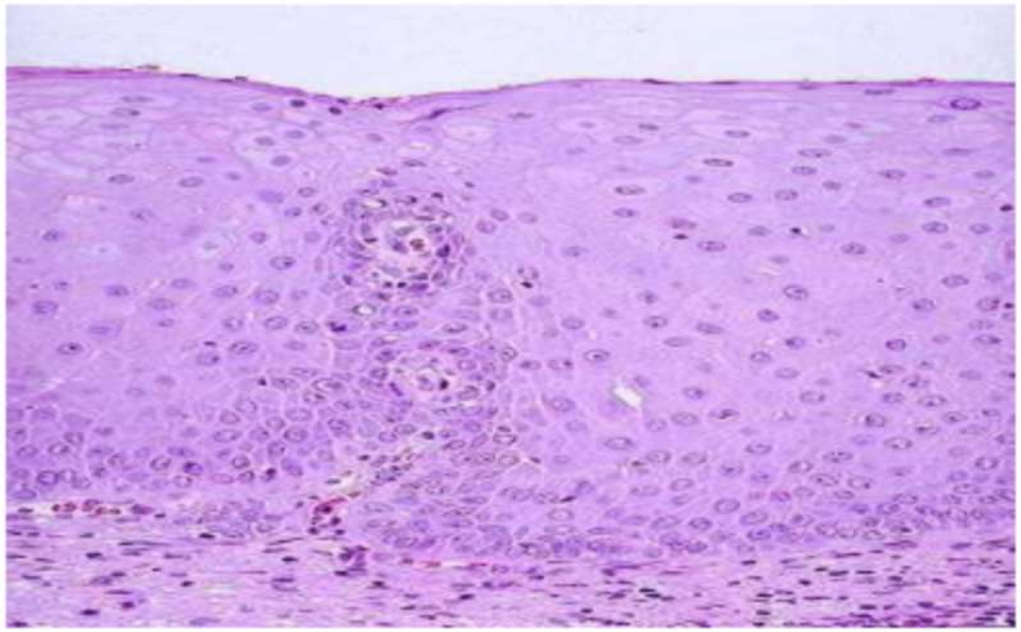
Normal squamous epithelium



Neoplasia: Well differentiated squamous cell carcinoma

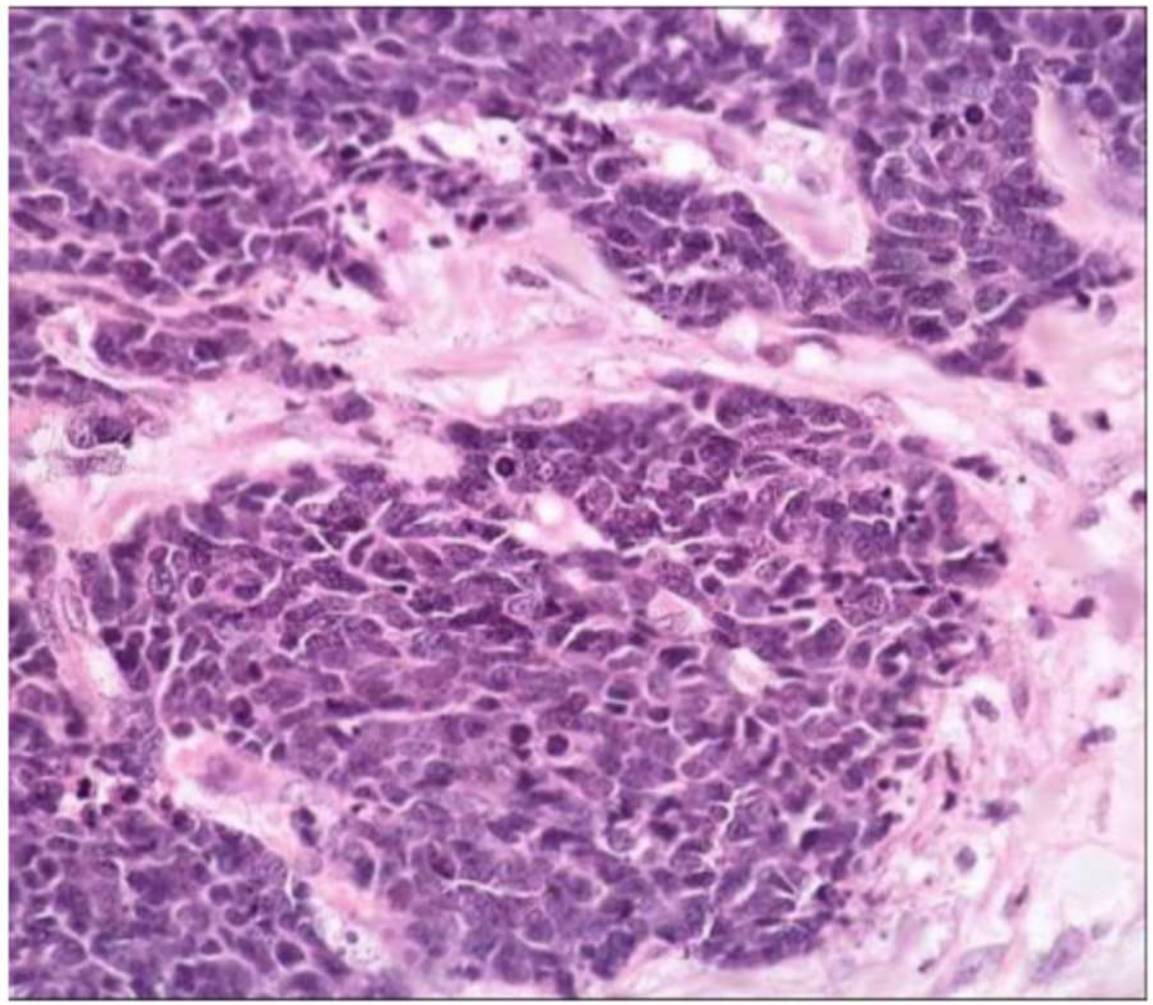


The neoplasia still forms squamous epithelium with identifiable keratin production, keratin pearls and intercellular bridges



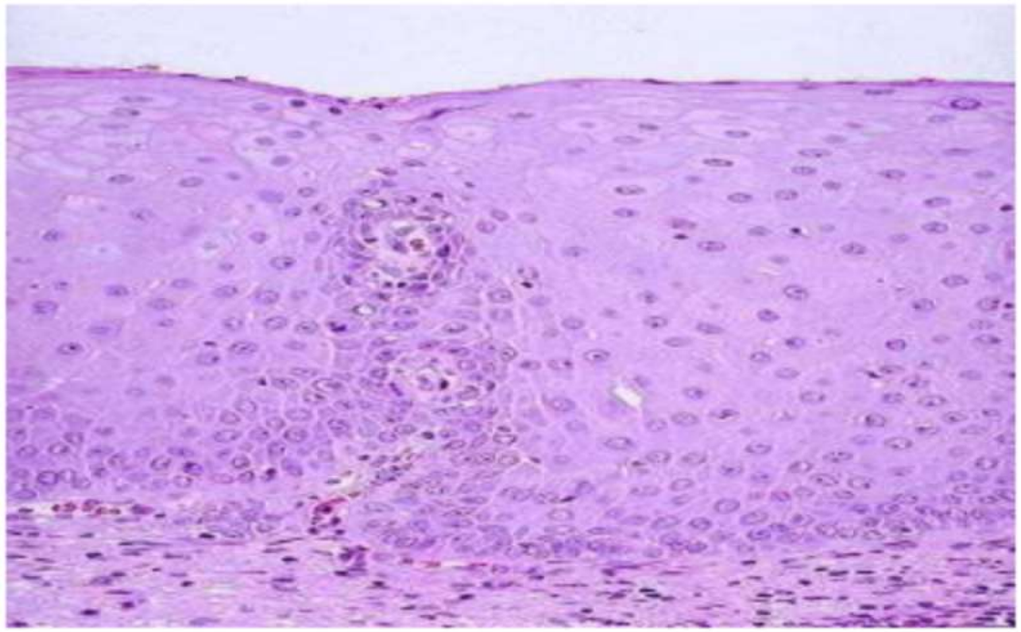


Neoplasia: Poorly differentiated carcinoma

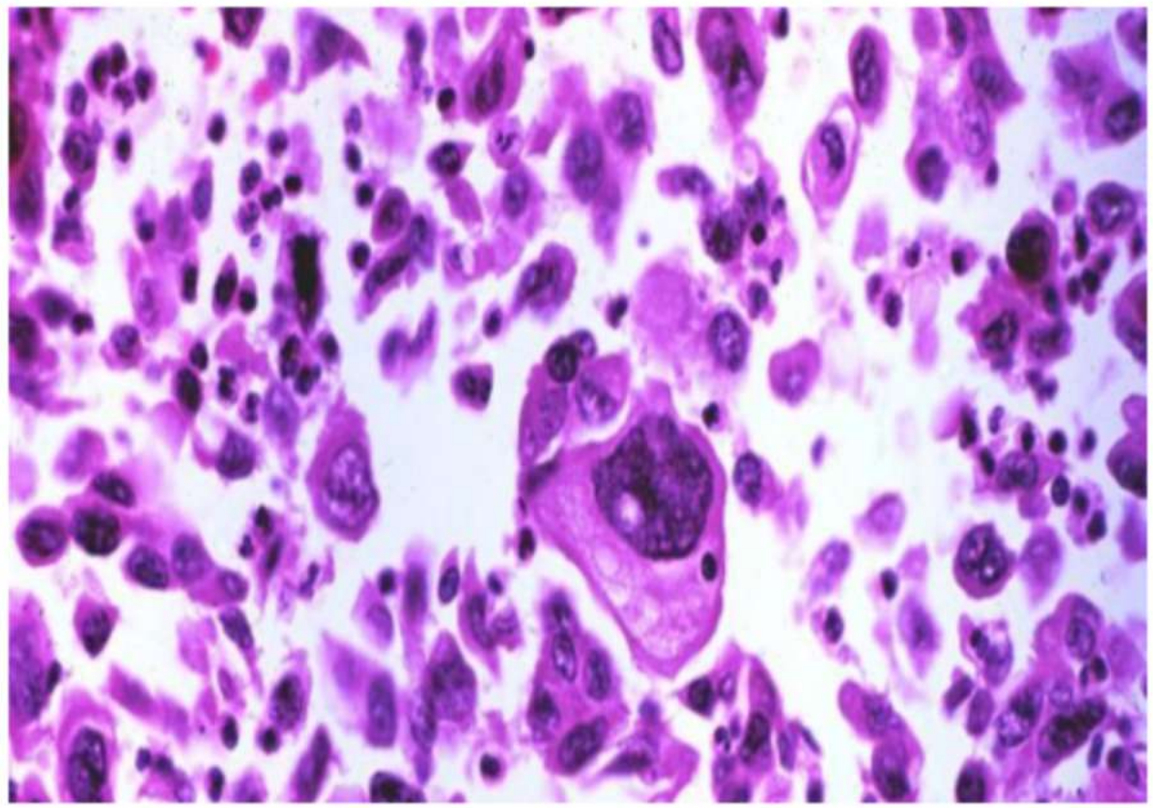


Look at the neoplasia: There is no identifiable structure that points towards glandular or squamous differentiation

Anaplasia is lack of differentiation







The cells are pleomorphic (variability in the size and shape of cells and/or their nuclei)

Differentiation and Anaplasia

- Function:
  - Benign neoplasm: have normal function eg. Adrenal cortical adenoma (still produces adrenal cortical hormone)
  - Malignant neoplasm: have variable function according to the degree of differentiation eg. Hepatocellular carcinoma produces bile

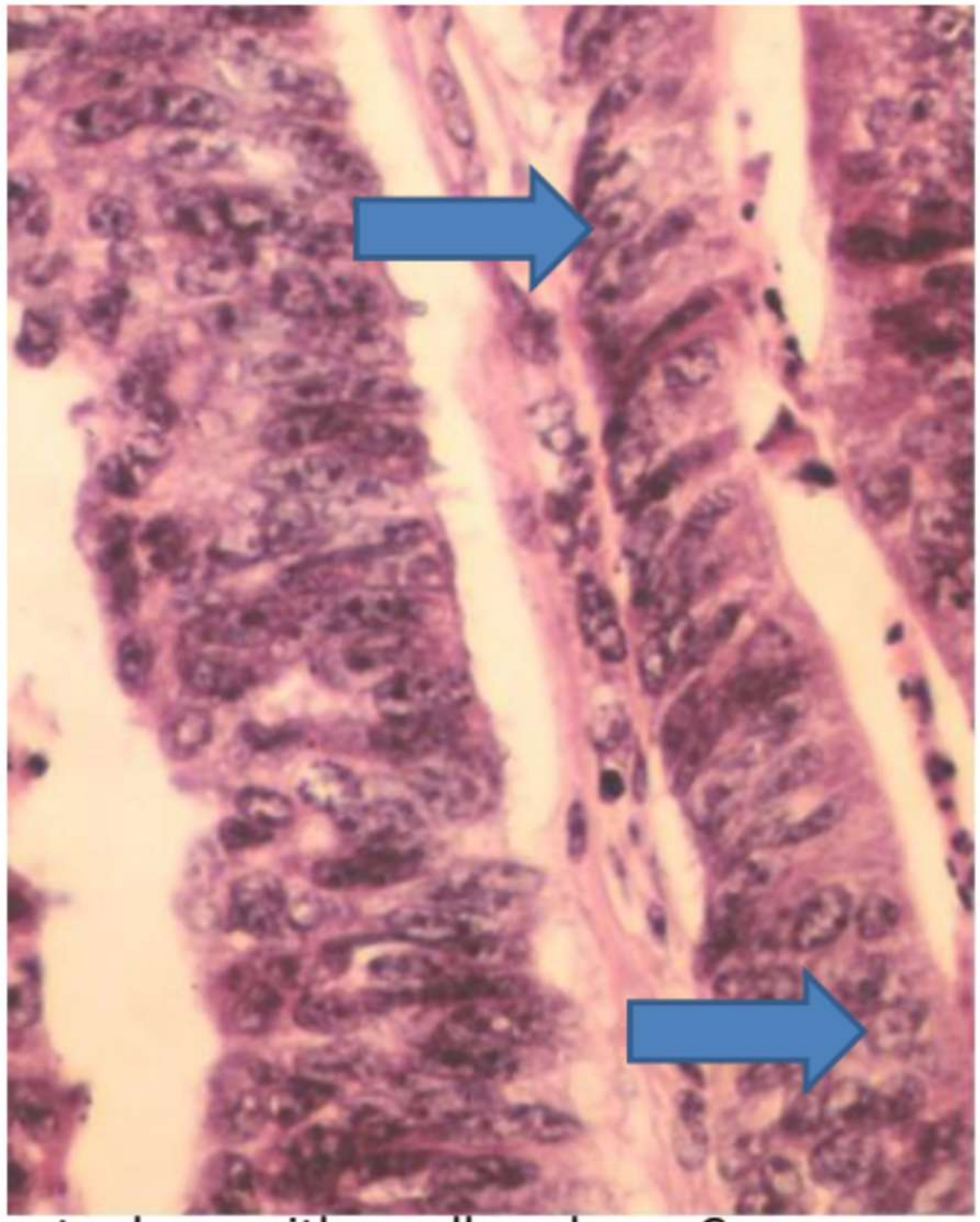
Nuclear features

- Benign neoplasm
  - Nuclear to cytoplasmic (N:C) ratio is close to normal
  - (Lots of cytoplasm vs small nucleus)
  - Mitosis have normal mitotic spindles
- Malignant neoplasm
  - Nuclear to cytoplasmic (N:C) ratio is increased and nucleoli are prominent
  - Mitoses have normal and atypical mitotic spindles

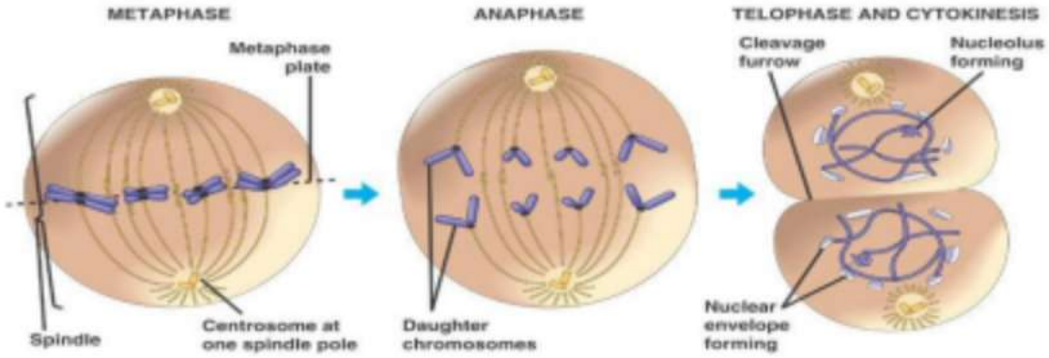
Normal NC ratio



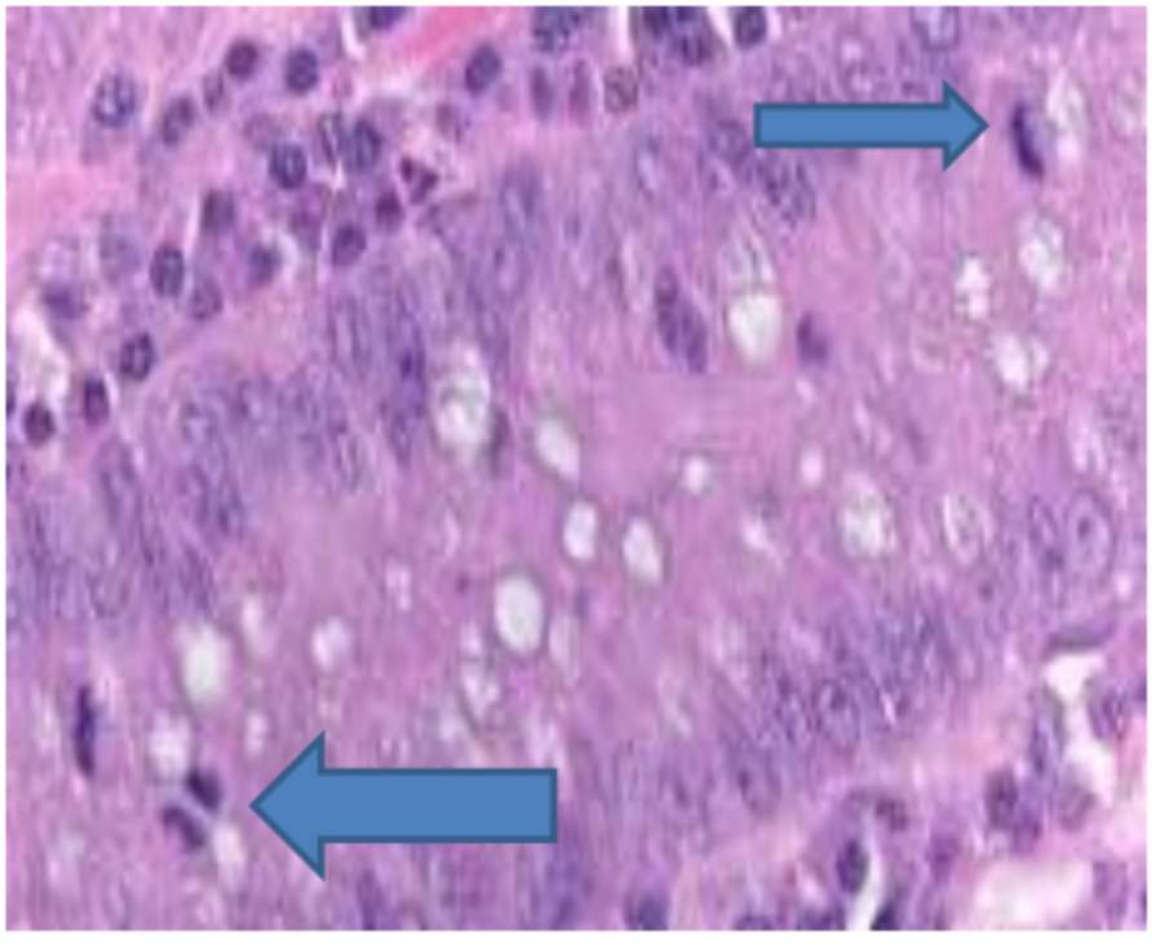
Increased NC ratio



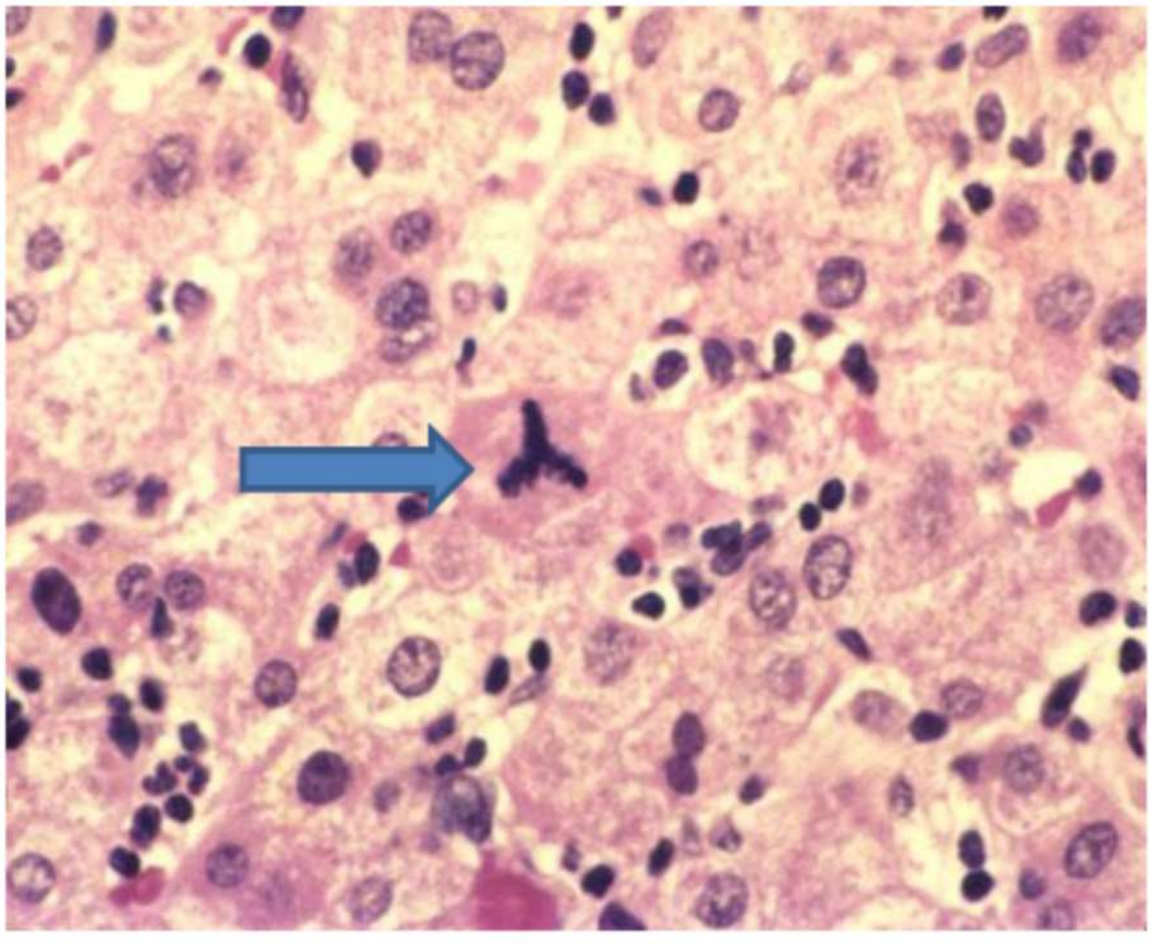
Look at the Normal cell. There is abundant cytoplasm with small nucleus. Compare that to the Malignant cell with increased NC ratio. The nucleus is large with reduced cytoplasm.



Typical mitosis



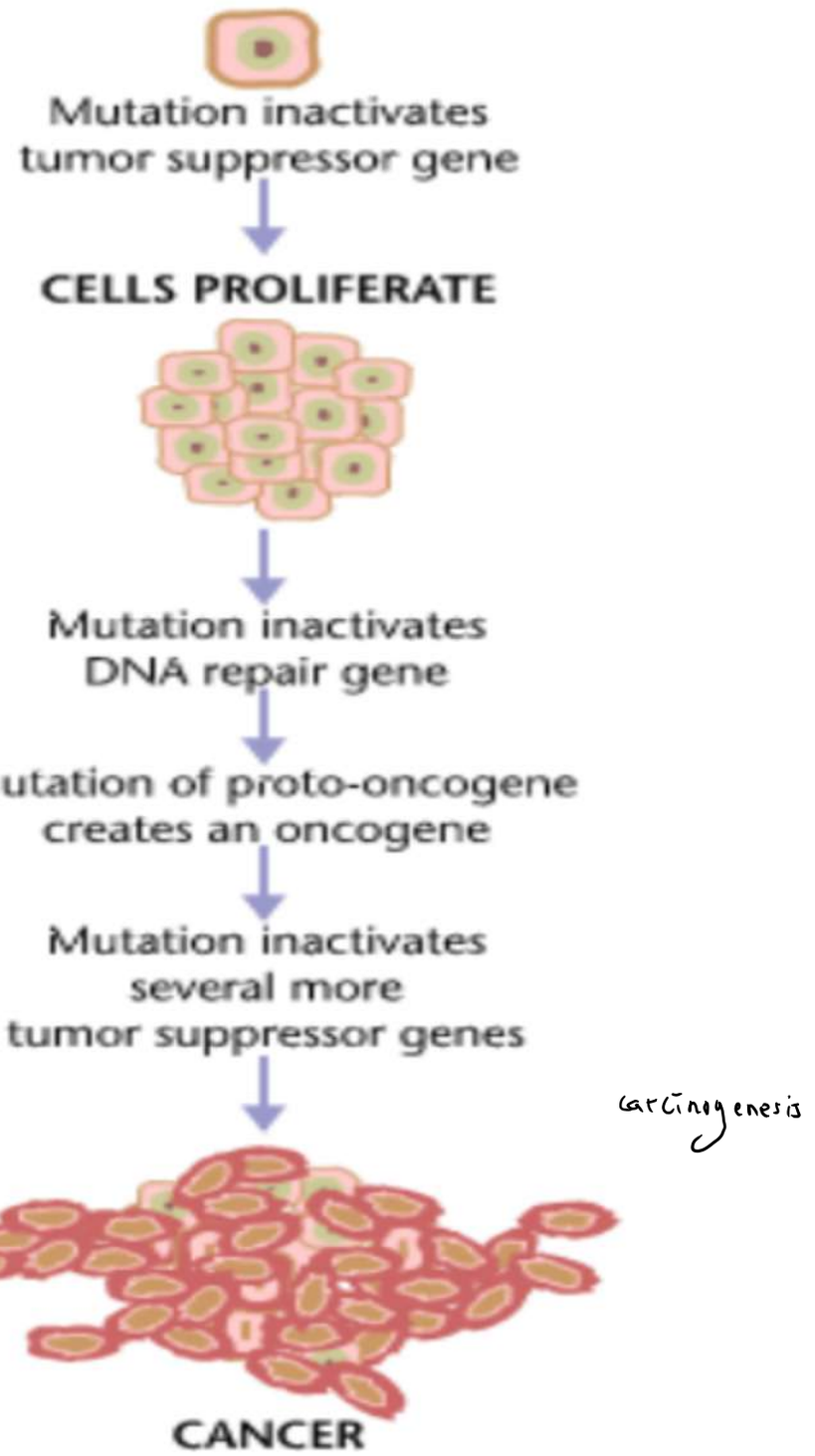
Atypical mitosis



Look at the typical mitosis. It can show metaphase, anaphase or telophase stage. Atypical mitosis show tripolar shape such as the “Mercedes benz” logo

Rate of Growth

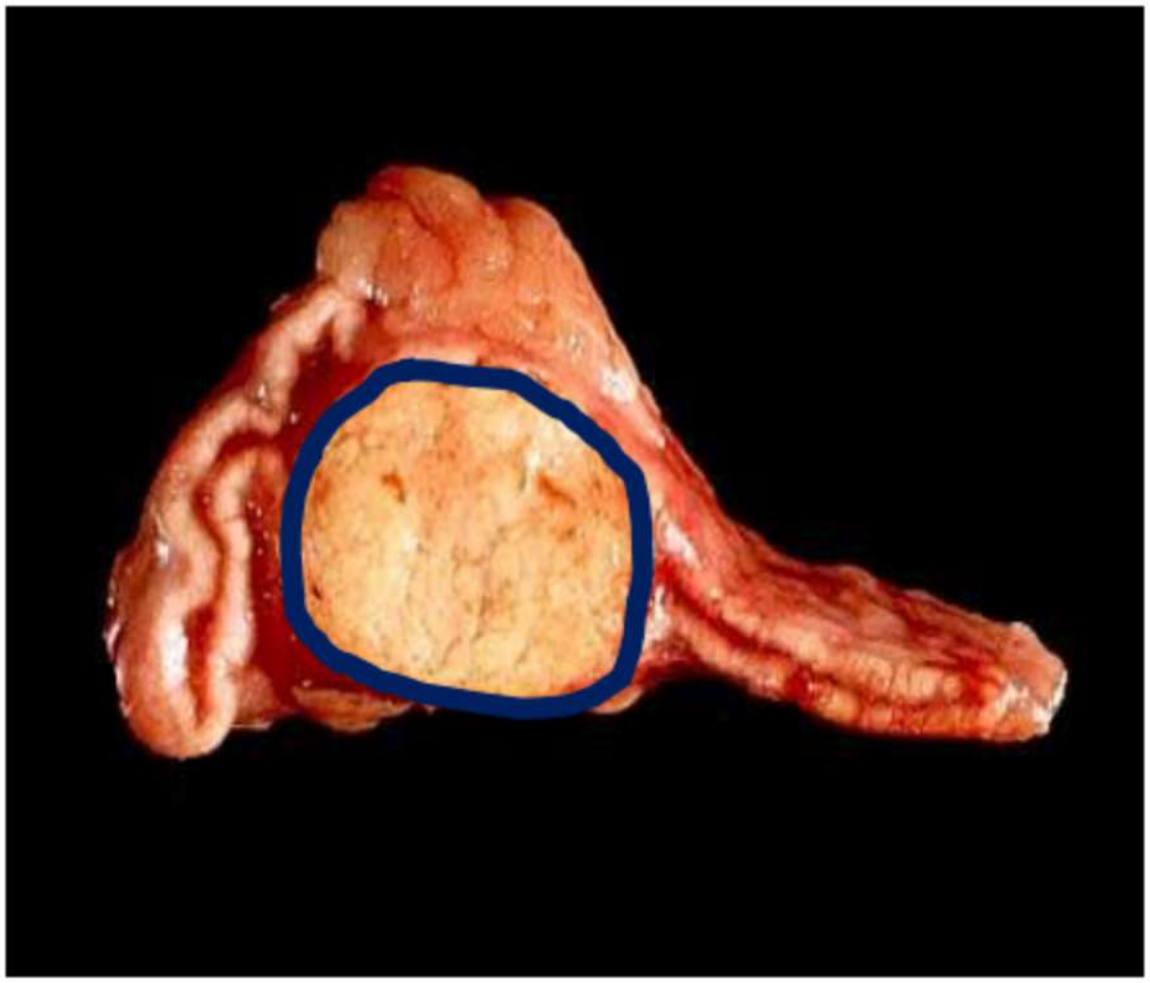
- Benign: usually is slow growing
- Malignant:
  - have a variable growth rate
  - usually is fast growing
- How long does it take to produce a clinically detectable neoplastic mass?
- estimated the transformed cell must undergo at least 30 population doublings to produce 10^9 cells, which is the smallest clinically detectable mass.



Local Invasion: Benign neoplasm

Pituitary adenoma
- benign - from corticotroph cells of APG
- divide into microadenoma & macroadenoma
- overproduce adrenocorticotropic hormone (ACTH) → cushing syndrome

- Do not invade
- they grow and expand slowly, usually develop a rim of compressed connective tissue, (fibrous *capsule)* which separates them from host tissue.
- capsule is derived largely from the extracellular matrix of the native tissue due to atrophy of normal parenchymal cells under pressure of an expanding tumor.
- keeps the benign neoplasm as a discrete, readily palpable, and easily movable mass that can be surgically enucleated

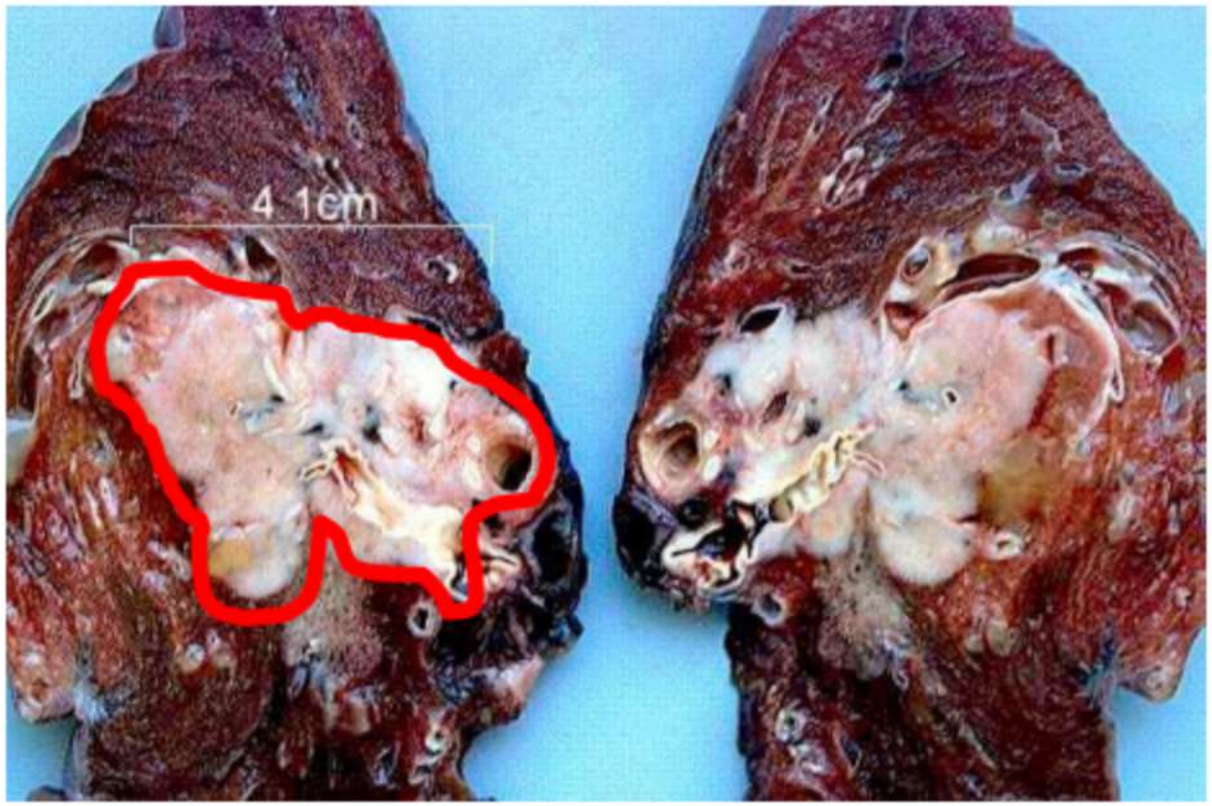


This is a well circumscribed tumour which do not invade the surrounding tissue. Adrenal adenoma

- Most adrenocortical carcinoma (ACC) arise de novo (directly as malignant tumor)
- An adrenocortical is benign, low risk of malignant transformation into ACC adenoma

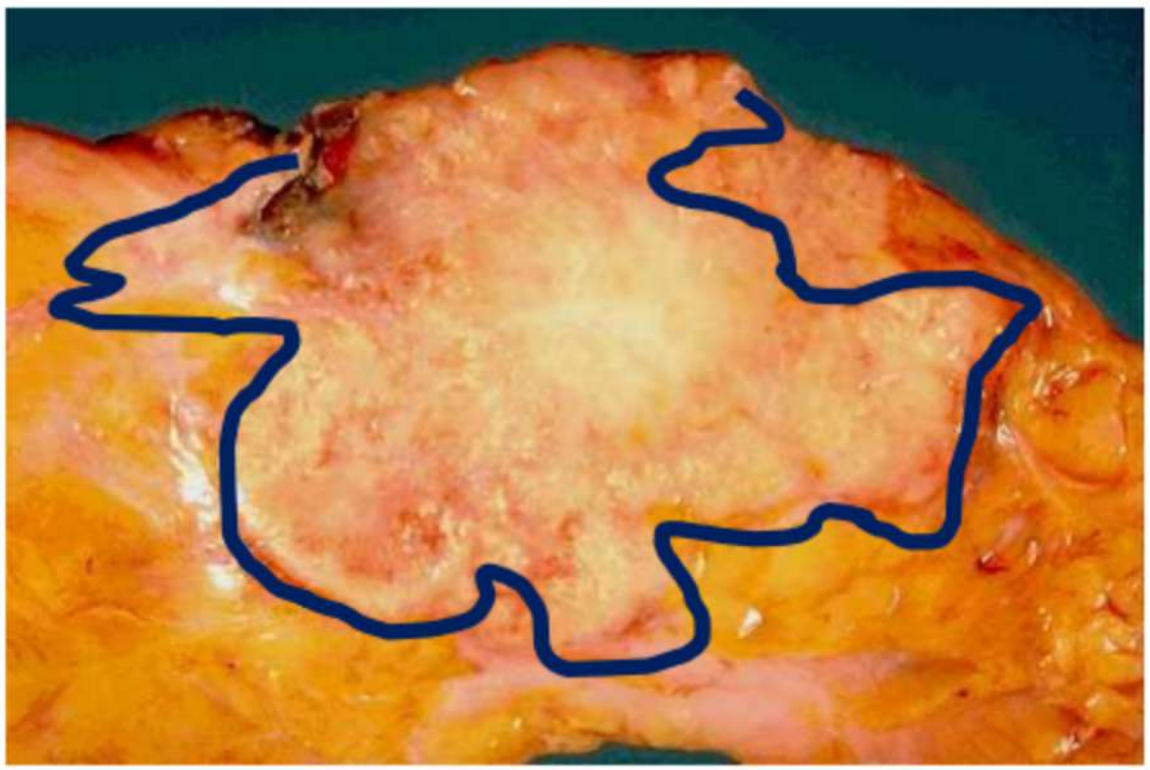
Local Invasion

Malignant: invasive, infiltrating, destructive, penetrating



Look at the invasive lung carcinoma. The tumour margin is irregular and not circumscribed compared to the benign neoplasia

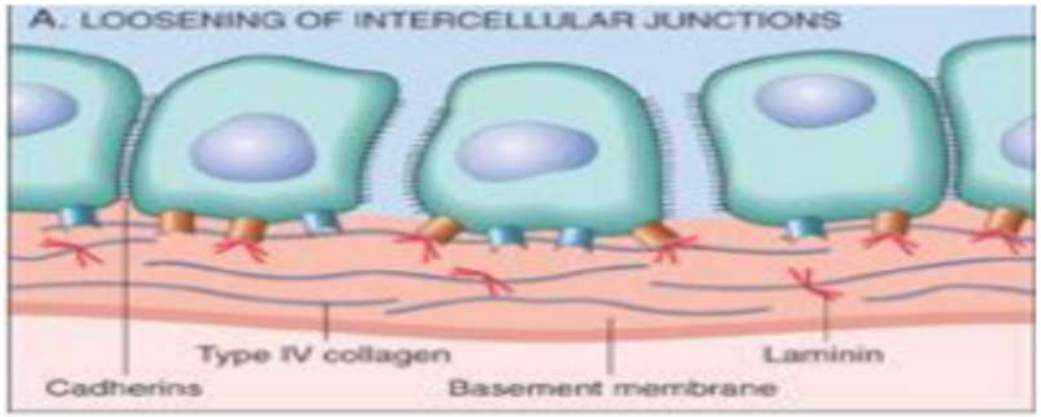
Invasion: second most important criterion for malignancy

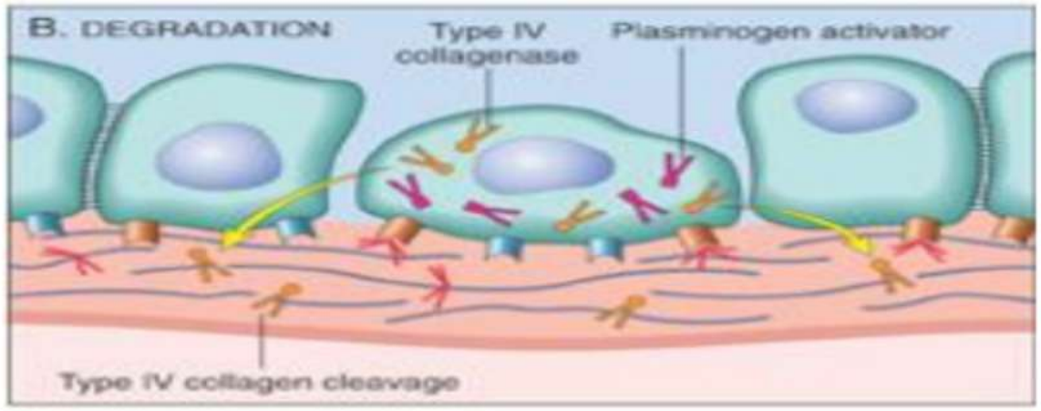


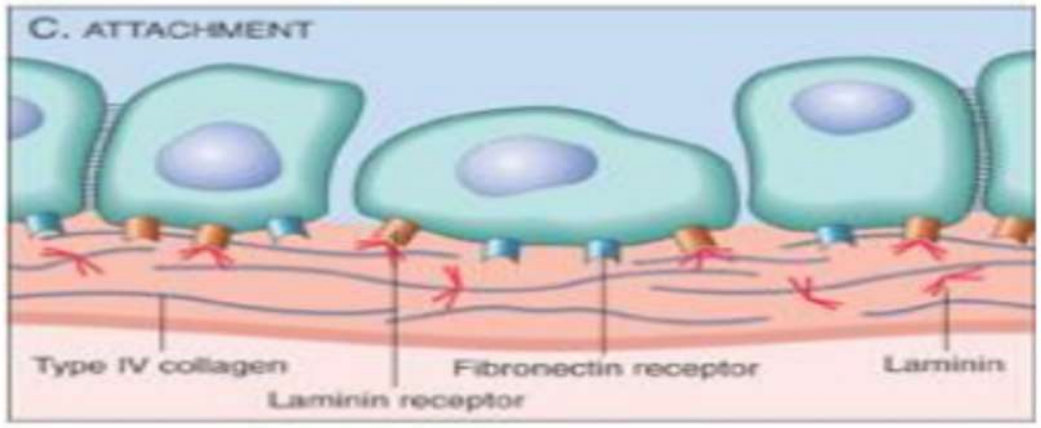
Another malignant adenocarcinoma showing invasion to the surrounding tissue

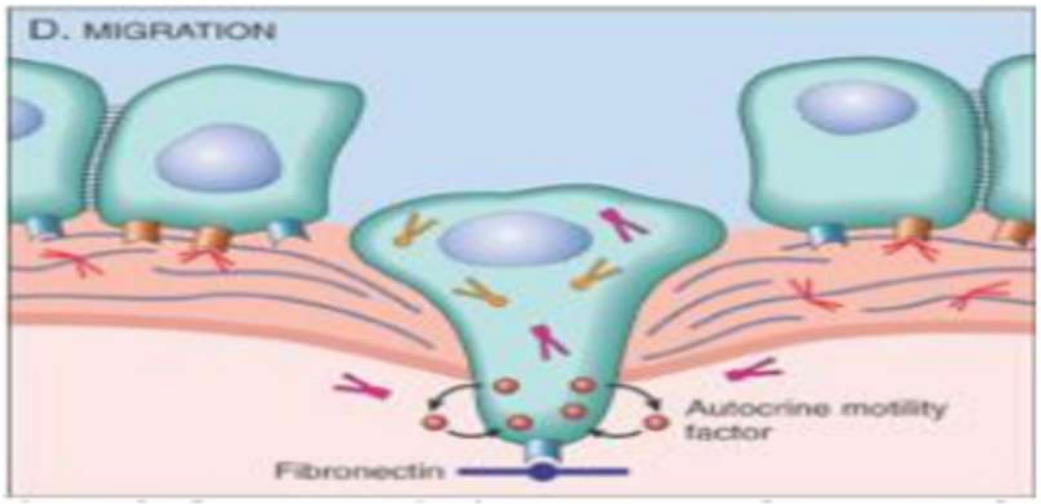
Sequence of invasion

- **Loss of intercellular adherence**
  - E-cadherin (intercellular adhesion agent) is *not* produced.
- Cell **receptors attach to laminin** (glycoprotein in the basement membrane).
- Cells **release type IV collagenase** (metalloproteinase containing zinc).
- Cell receptors **attach to fibronectin** in the extracellular matrix.
- Cells produce **cytokines (stimulate locomotion)** and proteases **(dissolve connective tissue).**
- Cells produce factors that **stimulate angiogenesis.**
- Secrete vascular endothelial growth factor and basic fibroblast growth factor









Metastasis

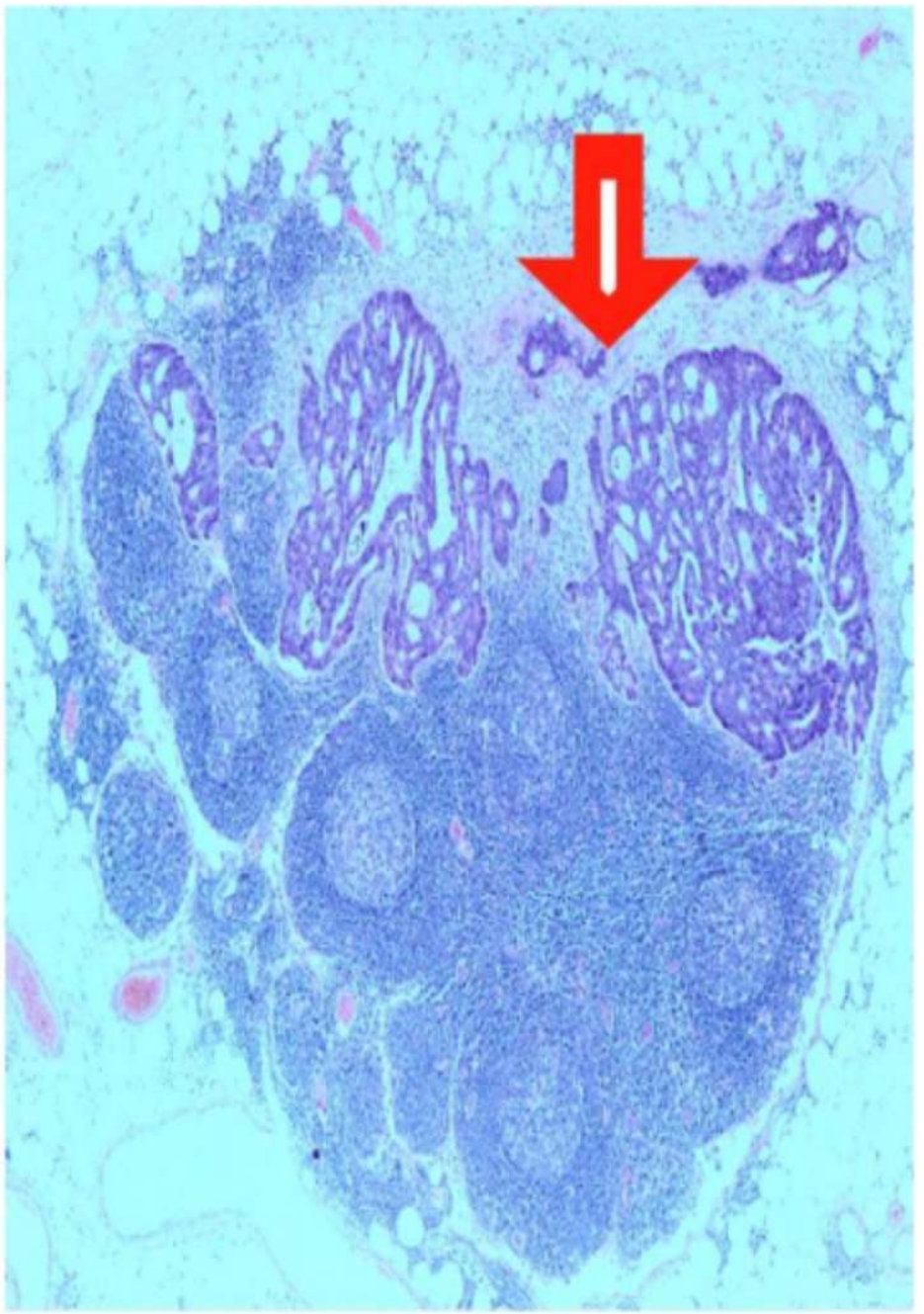
- The spread of cancer from one part of the body to another.
- Benign neoplasm do not metastasize.
- Malignant neoplasm metastasize.
- Pathways of metastasis:
  - 1. Seeding in body cavity
  - 2. Lymphatic spread to lymph nodes
  - 3. Hematogenous spread
- Important terminology:
  - Primary site of malignancy
  - Secondary / Secondaries / metastatic tumour
- **Metastasis is often more common than primary cancer in:**
- Lymph node, Lungs, Liver, Bone

Metastases: Lymphatic spread

- Usual mechanism of spread for carcinomas
- Regional lymph nodes are the first line of defense against the spread of carcinoma

cytotoxic T-cells, NK cells, macrophages

- However if the nodal architecture is destroyed, the malignant cells enter the efferent lymphatics which drains into the bloodstream
- In the bloodstream, malignant cells metastasize to distant organs (lung, liver, bone etc)

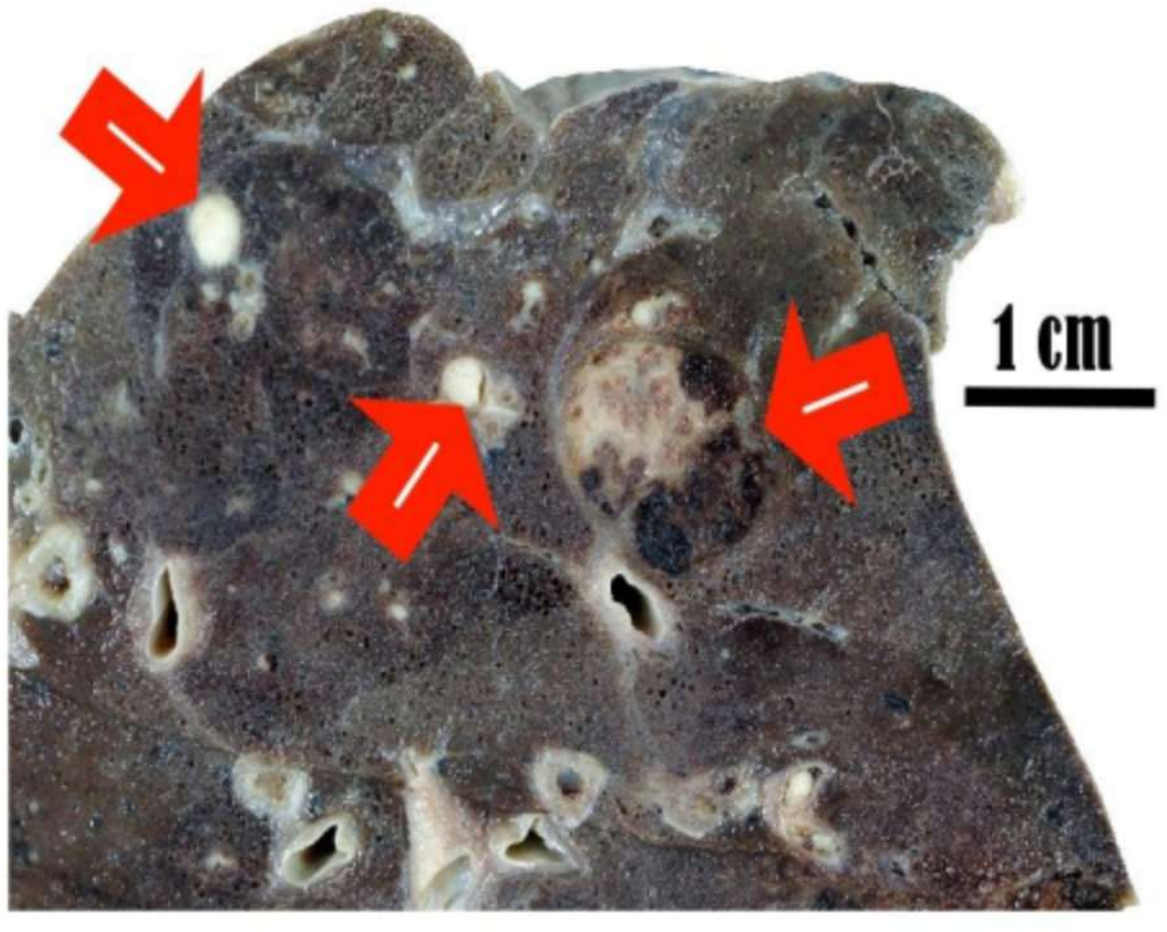


papillary thyroid carcinoma via lymphatics to cervical lymph nodes, rarely spreading hematogenously

Metastases: Hematogenous spread

hepatic encephalopathy

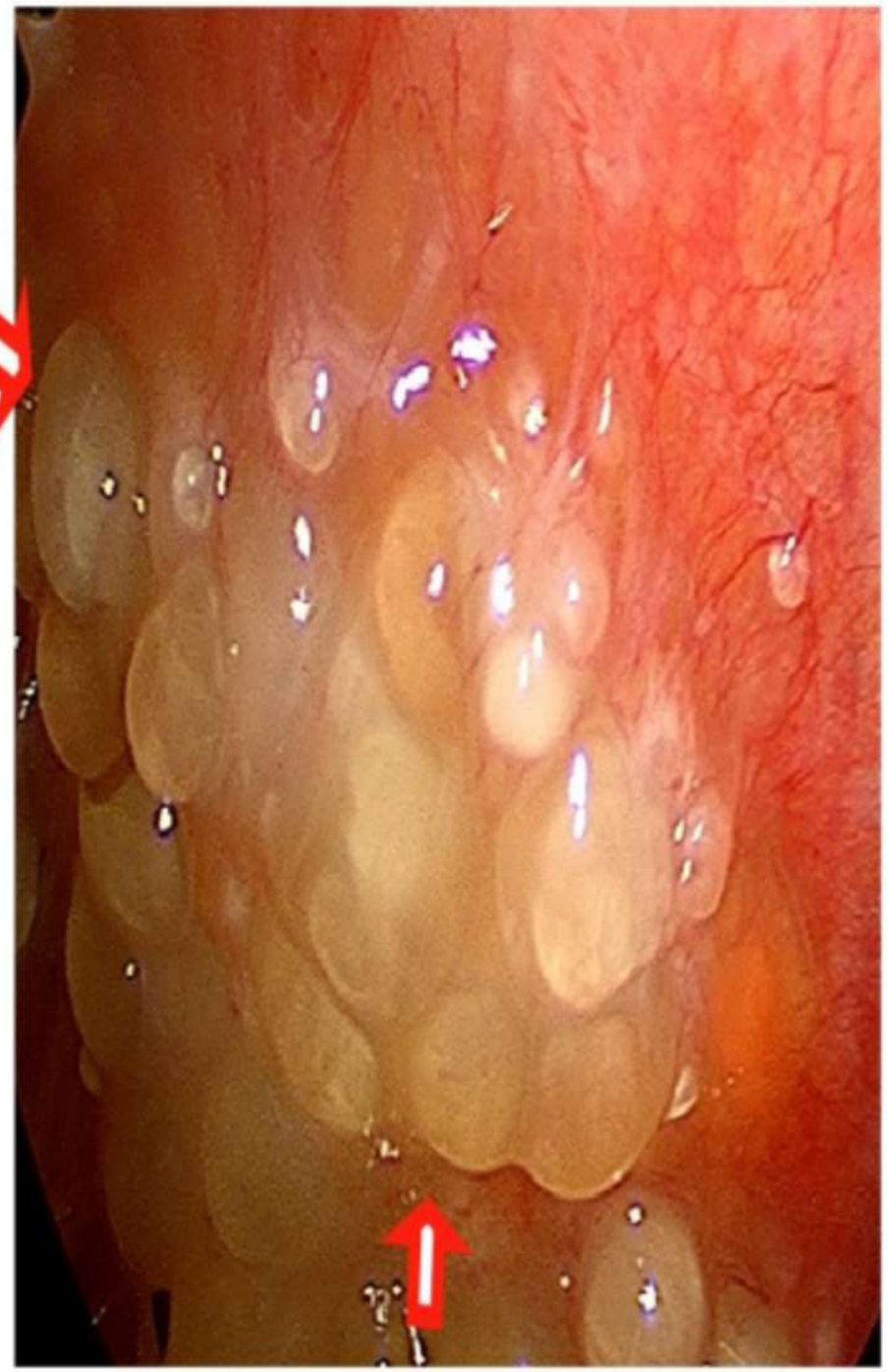
- Usual mechanism of spread for sarcoma
- Cells entering the portal vein metastasize to liver
- Cells entering vena cava metastasize to the lung
- Some malignancies have both lymphatics and hematogenous spread. Eg. Renal cell carcinoma and hepatocellular carcinoma
  - follicular thyroid ~
  - Choriocarcinoma



Metastasis: Seeding into body cavity

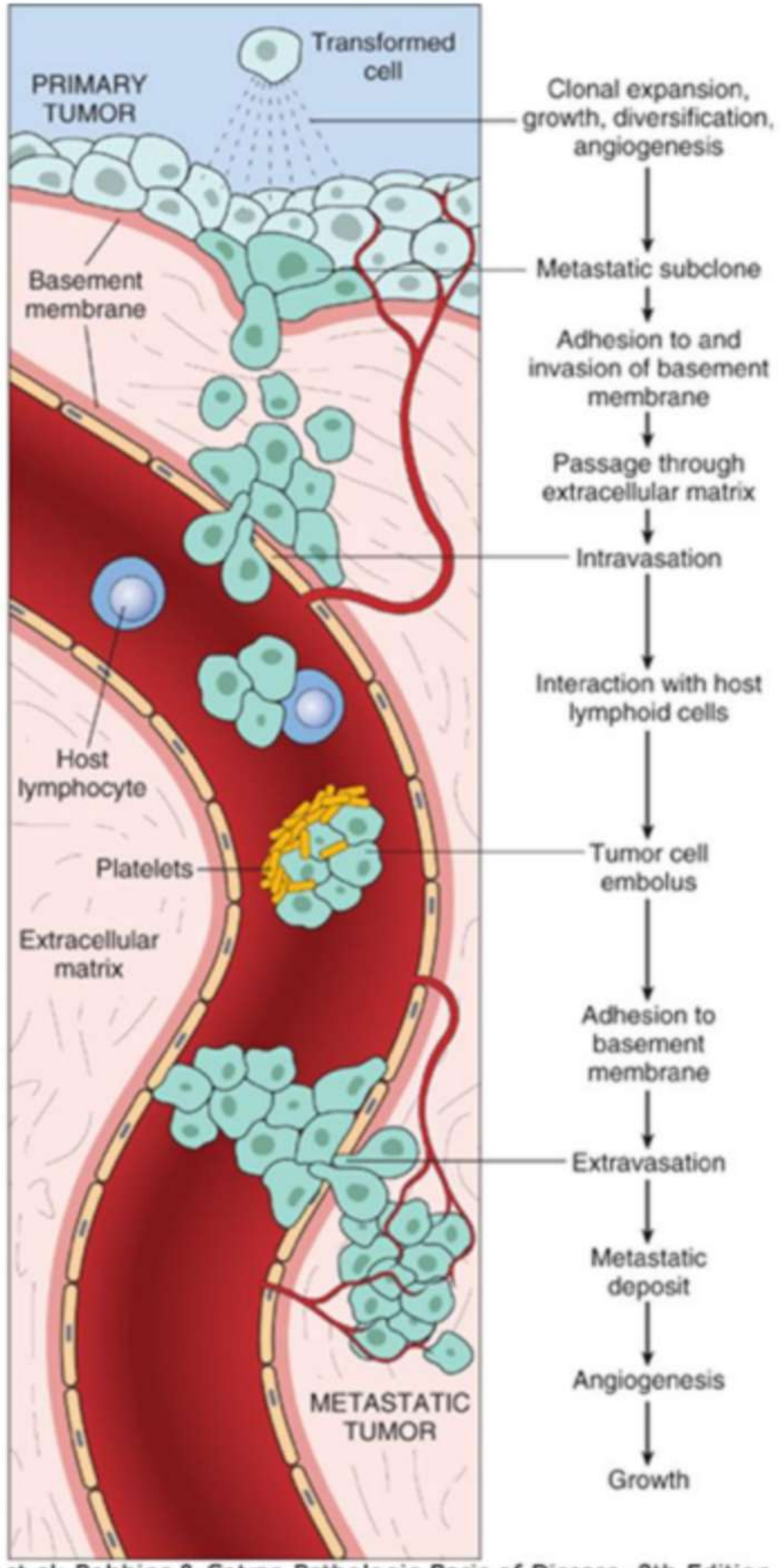
Seeding into body cavity

- Malignant cells exfoliate from a surface and implant and invade tissue in a body cavity.
- Primary surface-derived ovarian cancers (e.g., serous cystadenocarcinoma) commonly seed the omentum.
- Peripherally located lung cancers commonly seed the parietal and visceral pleurae.
- Glioblastoma multiforme commonly seeds the cerebrospinal fluid causing spread to the brain and spinal cord.



5 major steps in metastasis

1. Invade surrounding normal tissue with penetration of small lymphatic or vascular channels
2. Release of neoplastic cells, into the circulation
3. Survival in the circulation;
4. Stop in the capillary beds of distant organs;
5. Penetrate lymphatic or blood vessel walls followed by growth of the disseminated tumor cells

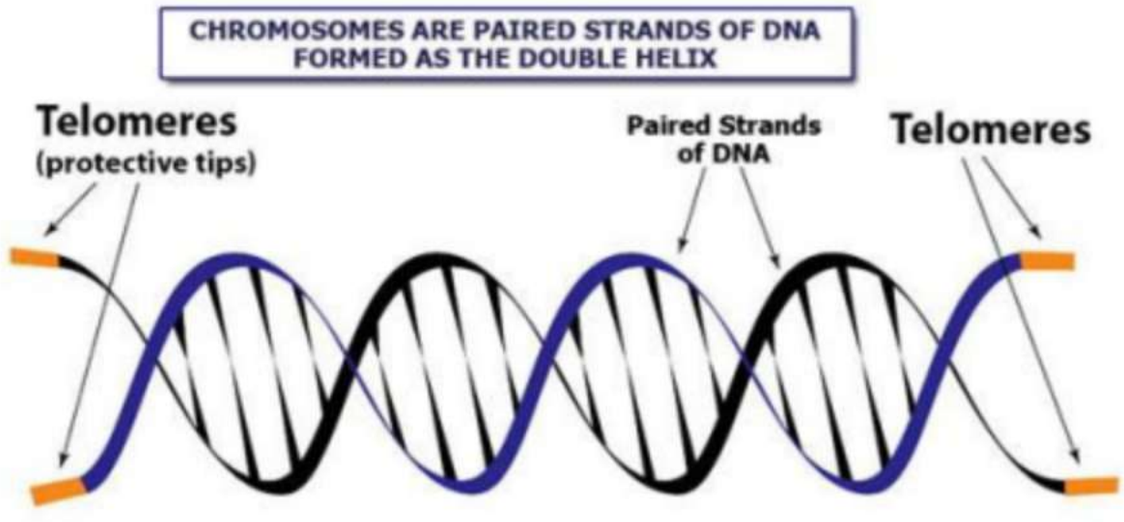


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Telomerase activity

What is a telomere?

- a compound structure at the end of a chromosome.
- It gets shorter with cell division
- Cells normally can divide only about 50 to 70 times, with telomeres getting progressively shorter
- So it is the cells' internal clock to count the number of cell divisions that has been made and stops further divisions after a set number



Telomerase activity

What is telomerase?

- Enzyme telomerase adds bases to the ends of telomeres.
- In young cells, telomerase keeps telomeres from wearing down too much.
- Telomerase remains active in sperm and ovum

Telomerase and neoplasia

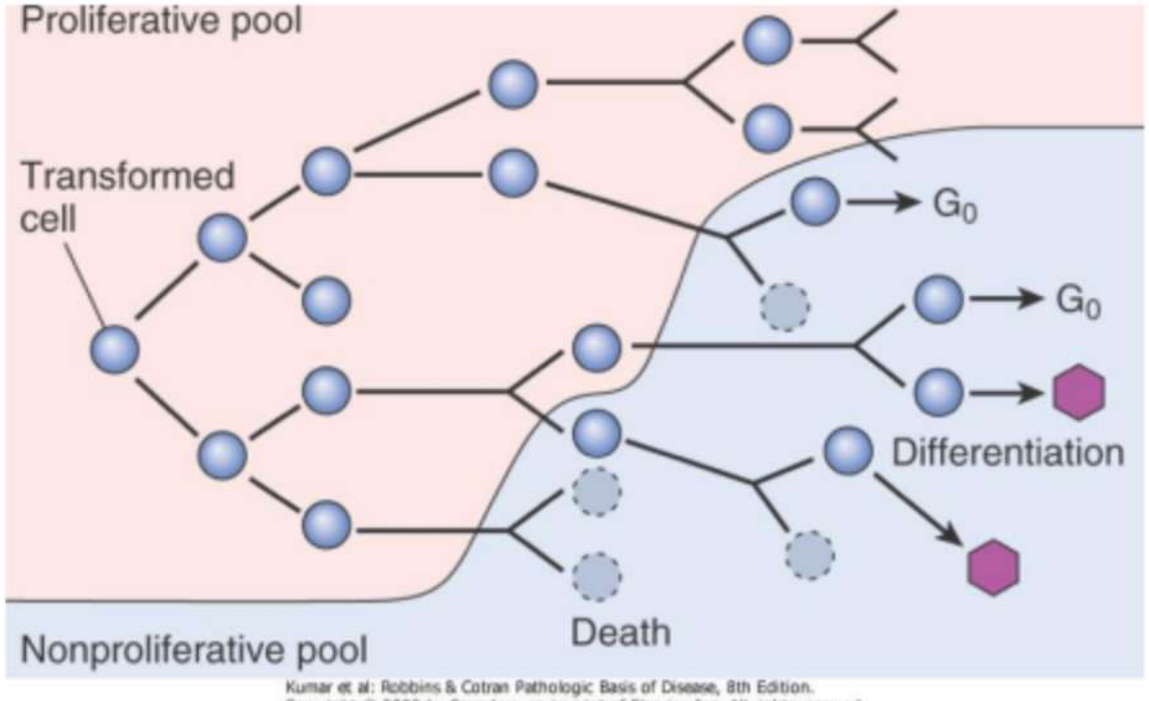
- As a cell become cancerous, it divides more often, and its telomeres become very short.
- If its telomeres get too short, the cell may die. (replicative senescence & mitotic crisis)
- Cancer cell can activate telomerase, which prevents the telomeres from getting even shorter.

Telomerase activity

- Benign neoplasm have normal telomerase activity.
- Malignant neoplasm have upregulation of telomerase activity.
- They do *not* lose genetic material after multiple cell divisions.

Monoclonality

- Benign and malignant neoplasms derive from a single precursor cell (monoclonal)
- Non-neoplastic proliferations derive from multiple cells (polyclonal).



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Benign and Malignant

- What are the characteristics of benign and malignant neoplasm?
- Features:
  - Differentiation and Anaplasia
  - Nuclear features
  - Rate of Growth
  - Local Invasion
  - Metastasis
  - Telomerase activity
  - Monoclonality

Normal

Premalignant

(Benign neoplasia /Dysplasia)

malignant

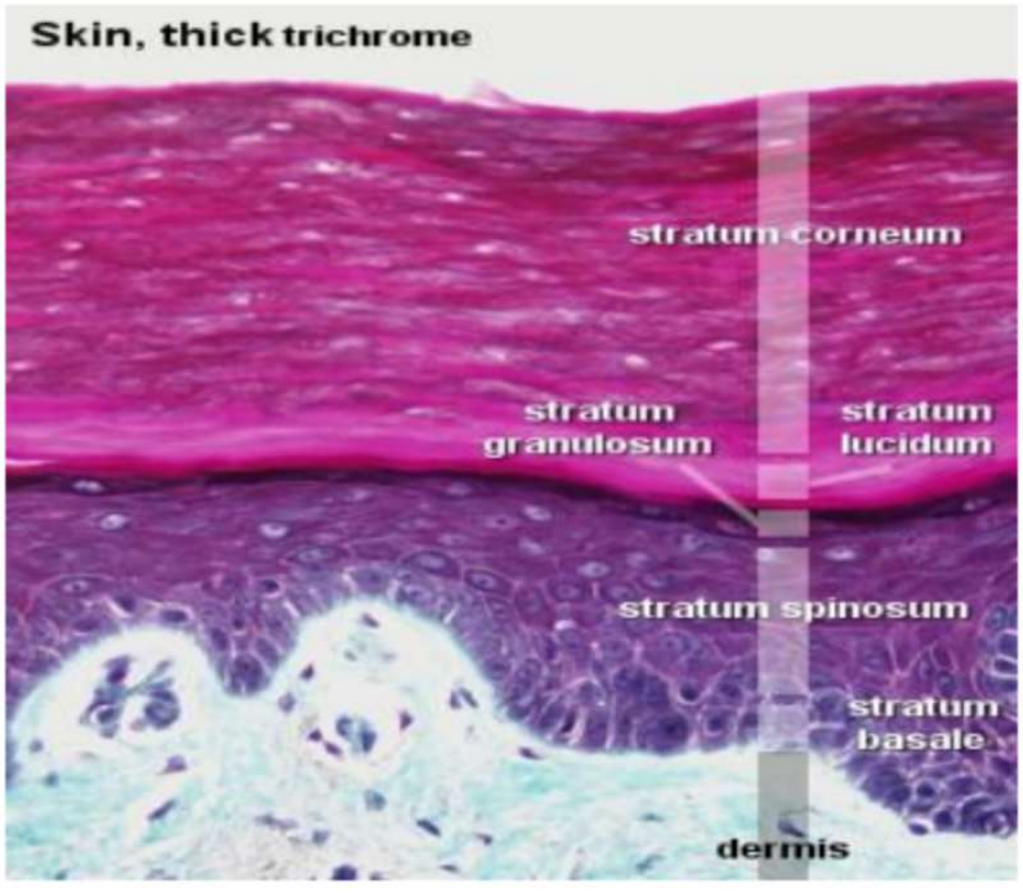
Part 3: Dysplasia

if severe = carcinoma in situ

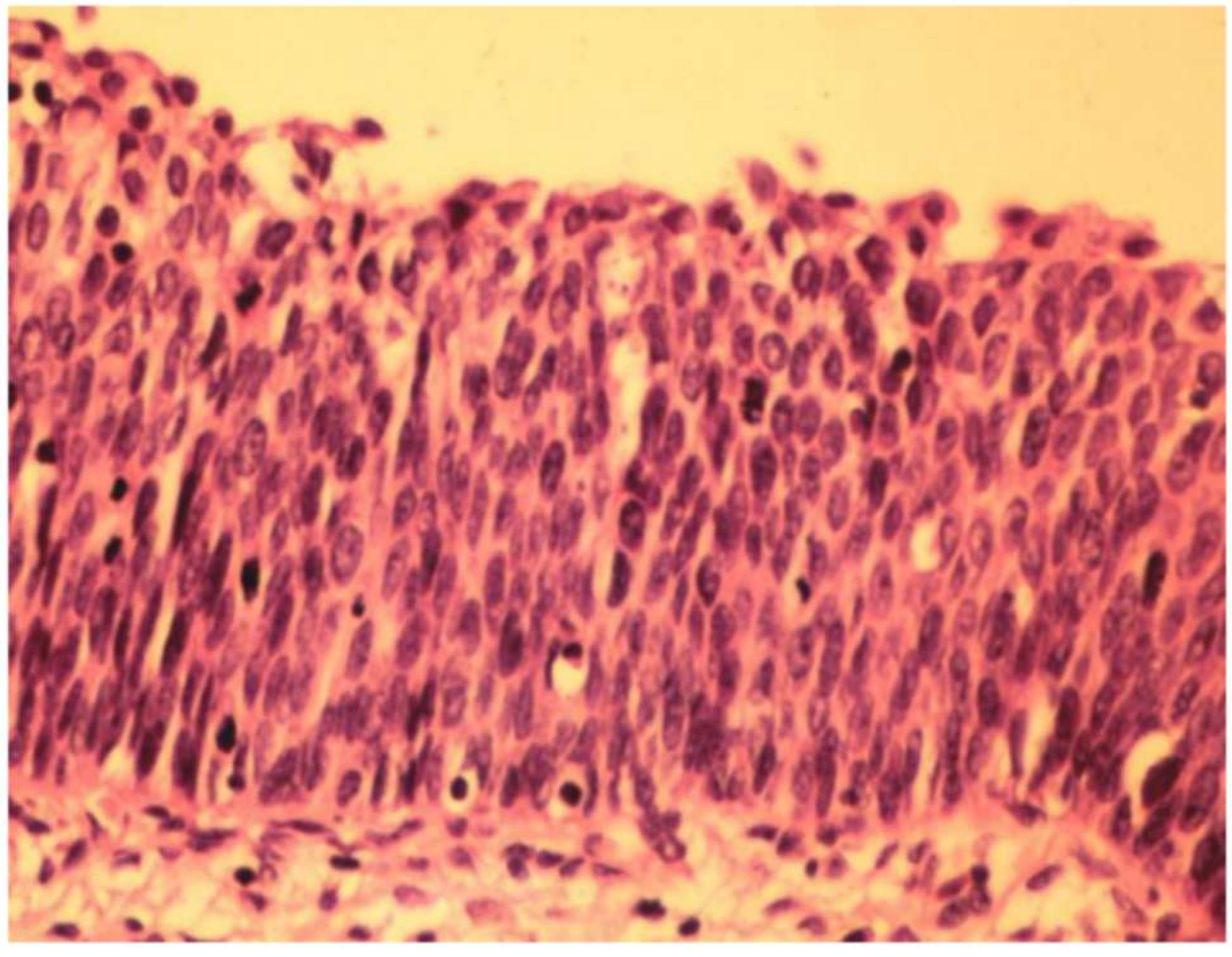
Ex: - cervical dysplaxia
- barett's esophagus (esophageal dysplaxia)
- tubular adenomas of colon

- Not neoplastic growth
- loss of architectural organization and a loss of cell uniformity in epithelium
- pleomorphism and mitoses are more prominent than in the normal
- usually graded: mild, moderate, severe, and carcinoma-in-situ

normal epithelium



dysplasia



The left picture shows normal squamous epithelium with definitive architectural organization and uniformity. Note that there is no pleomorphism and mitosis are only found at the stratum basale. The other picture shows loss of the normal architectural organization and uniformity with cellular pleomorphism and increased mitosis throughout the epithelium. These are the features of dysplasia

Dysplasia

- Dysplasia is a non-neoplastic proliferation.
- It is important to recognize that dysplasia often does not progress to malignancy
  - It may be reversible, especially when low grade
- It is not possible to predict from morphology which cases will progress and which will not
  - Other markers can be helpful
    - Ex. HPV testing in cervical dysplasia

detects presence of high-risk human papillomavirus (hrHPV) DNA or RNA in cervical cells, identifying underlying viral infection that causes nearly all cervical dysplasia and cancer

Acquired preneoplastic disorders

PRECURSOR LESION	CANCER
Actinic (solar) keratosis	Squamous cell carcinoma
Atypical hyperplasia of ductal epithelium of breast	Adenocarcinoma
Chronic irritation at sinus orifice, third-degree burn scars	Squamous cell carcinoma
Chronic ulcerative colitis	Adenocarcinoma
Complete hydatidiform mole	Choriocarcinoma
Dysplastic nevus	Malignant melanoma
Endometrial hyperplasia	Adenocarcinoma
Glandular metaplasia of esophagus (Barrett's esophagus)	Adenocarcinoma
Glandular metaplasia of stomach (*Helicobacter pylori*)	Adenocarcinoma

Part 4: Cancer Incidence

- Cancers in children
  - Second most common cause of death in children (accidents most common cause)
- Most common cancer in children: acute lymphoblastic leukemia
  - Acute lymphoblastic leukemia (~33%), medulloblastoma (~21%), neuroblastoma (~7%), Wilms' tumor (~5%)
    - These are *not* common tumors in adults.

nephroblastoma

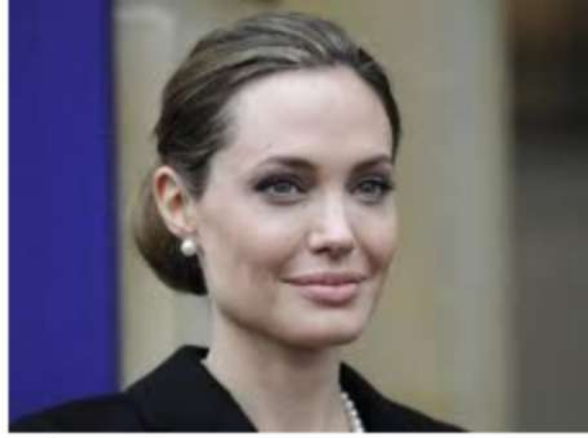
Cancer Incidence

- Cancers in men (in decreasing order)
  - Prostate, lung, colorectal
- Cancers in women (in decreasing order)
  - Breast, lung, colorectal
- Gynecologic cancers (in descending order)
  - Endometrium
  - Ovarian
  - Cervical (Least common due to cervical Pap smears detecting dysplasia)

Cancer related death

- Most common cause of cancer death in adults: lung cancer
- Cancer-related deaths in men (in decreasing order)
  - Lung, prostate, colorectal
- Cancer-related deaths in women (in decreasing order)
  - Lung, breast, colorectal
- Gynecologic cancer-related deaths (in descending order)
  - Ovary
  - Endometrium
  - Cervix

Cancer and heredity



- Inherited predisposition to cancer accounts for 5% of all cancers.
- Categories of inherited cancers
  - Autosomal dominant cancer syndromes
    - Retinoblastoma, Familial adenomatous polyposis, Li Fraumeni syndrome, BRCA1 and BRCA2 genes
  - Autosomal recessive disorders involving DNA repair
    - Xeroderma pigmentosum
- Familial cancers
    - No defined pattern of inheritance, but cancers (e.g., breast, ovary, colon) develop with increased frequency in families; sometimes involves *BRCA1* and *BRCA2* genes

CARCINOGENIC AGENTS

- **agent with the capacity to cause cancer in humans**
- **Chemical**
  - Direct and indirect acting
  - Initiation-promotion-progression
  - Eg Asbestos, Benzene
- **Micro-organisms**
  - Virus, bacteria and parasites
- **Radiation**
  - Ionizing radiation and UV light
- **Physical injury**
  - Burns and chronic inflammation



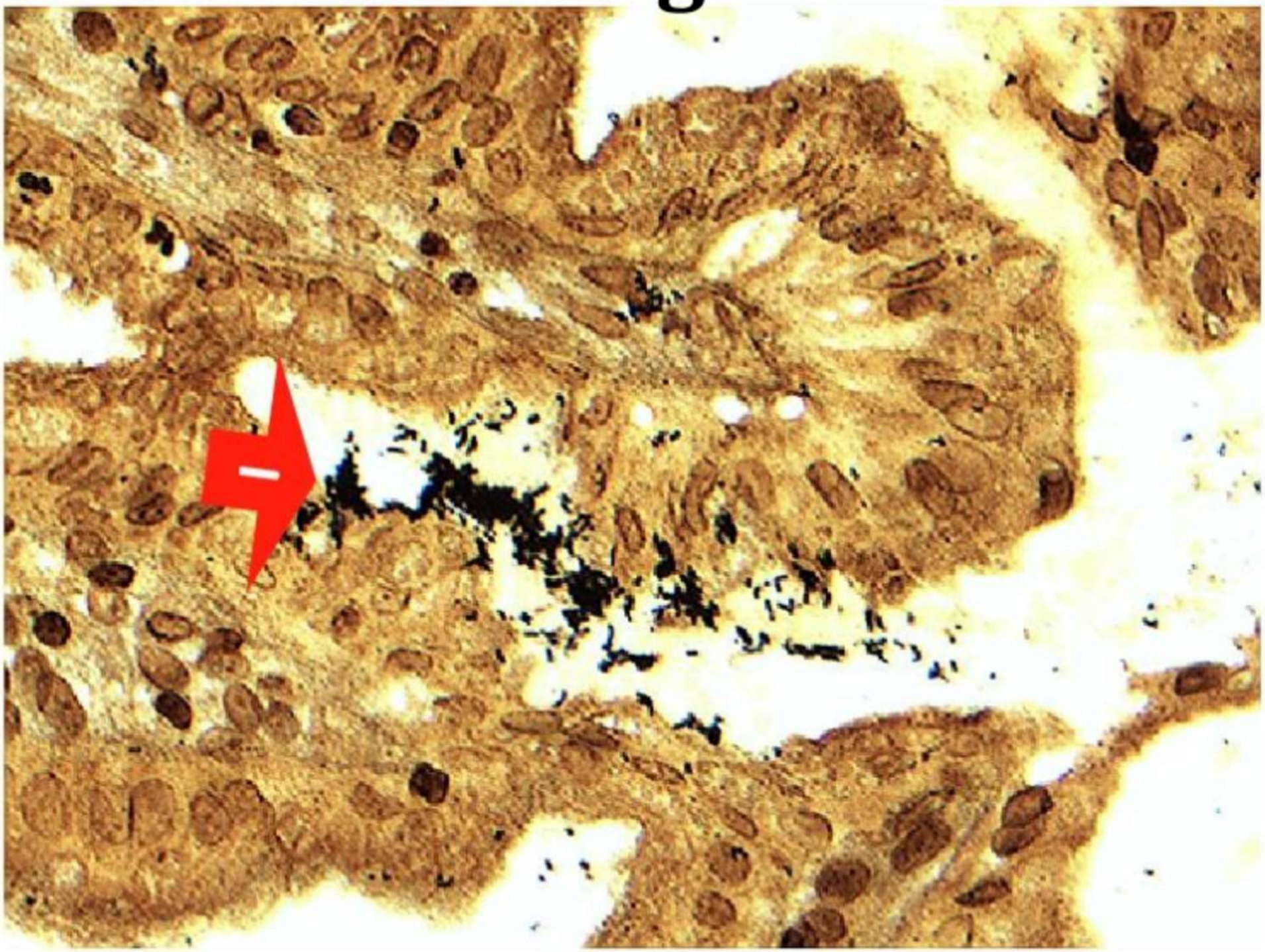
Cancer and geography

- Worldwide
  - Malignant melanoma: most rapidly increasing cancer
- Malaysia
  - Nasopharyngeal carcinoma secondary to Epstein-Barr virus (EBV)

positive test

- Japan
  - Stomach adenocarcinoma due to smoked foods
- Southeast Asia
  - Hepatocellular carcinoma due to hepatitis B virus plus aflatoxins (produced by *Aspergillus*) in food
- Africa
  - Burkitt's lymphoma due to EBV and Kaposi's sarcoma due to human herpesvirus 8

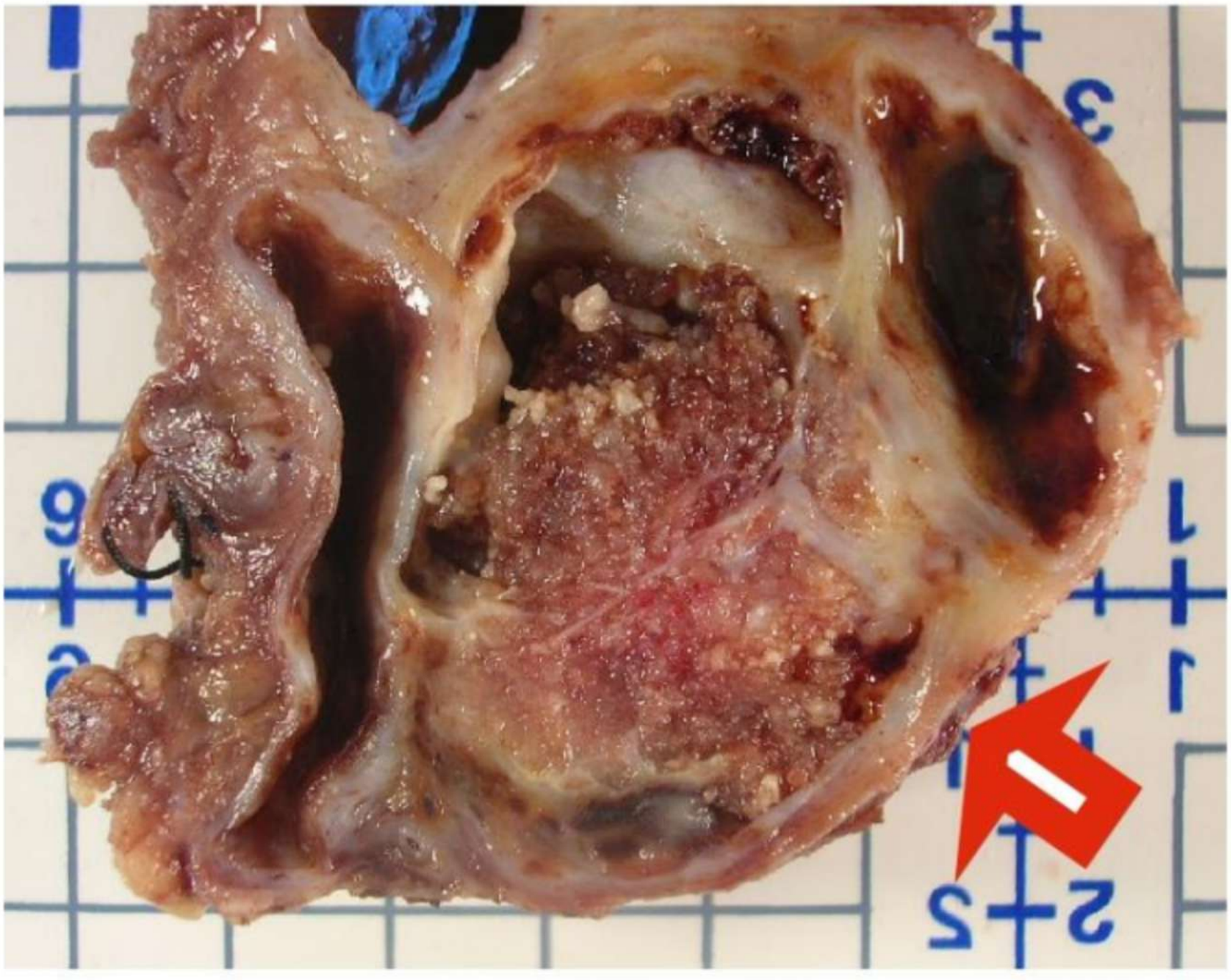
HELICOBACTER PYLORI
associated with gastric cancer



Liver carcinoma asscoiated with hepatitis B virus



THYROID CARCINOMA DUE TO RADIATION



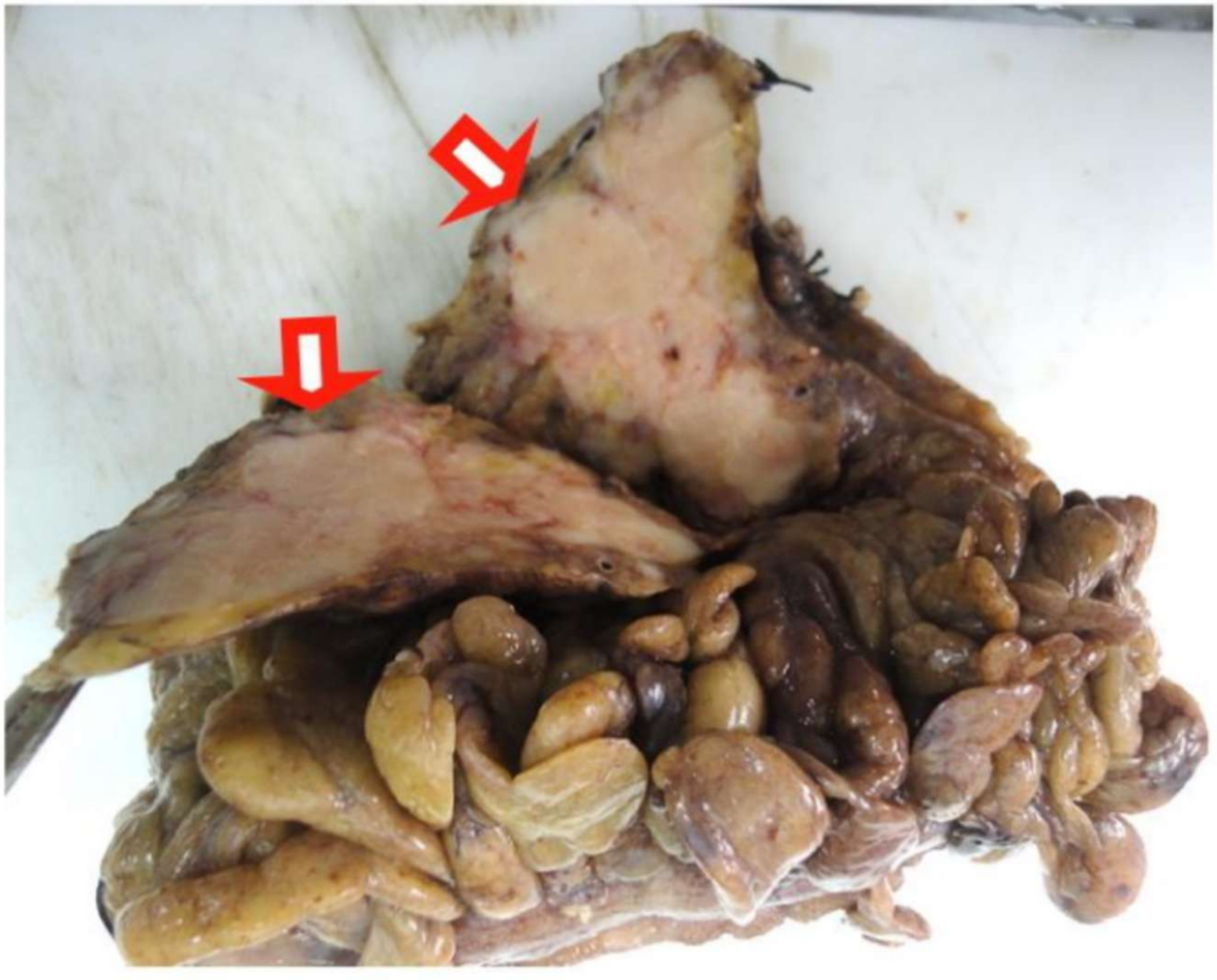
Part 5: NEOPLASIA SPREAD AND EFFECTS

CLINICAL EFFECTS OF TUMOURS : Local Effects

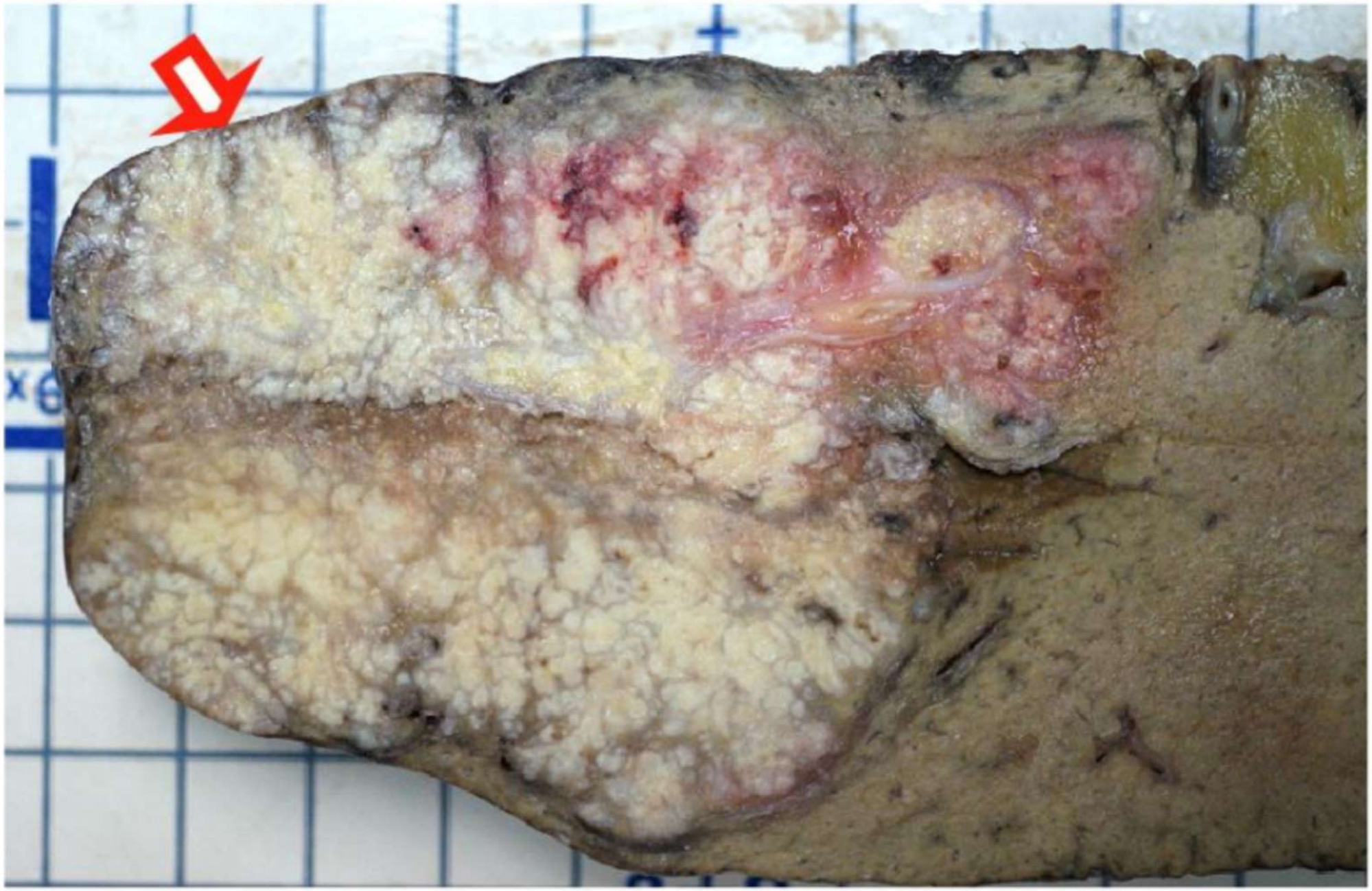
- Compression / obstruction
- Invasion
- Ulceration
- Destruction of adjacent organ
- Haemorrhage
- Infection
- Metastasis



COLONIC CARCINOMA & LYMPH NODE METASTASIS (DUKES C)



COLORECTAL CARCINOMA WITH LIVER METASTASIS (DUKES D)



CLINICAL EFFECTS OF TUMOURS

:METABOLIC EFFECTS

- **TUMOUR-TYPE SPECIFIC**
  - May retain the functional property
  - Especially well-differentiated endocrine neoplasm secreting hormones
- **NON SPECIFIC EFFECTS**
  - CACHEXIA (wasting syndrome) atrophy
  - •anorexia, muscle wasting, loss of subcutaneous fat
  - • due to tumour necrosis factor-ά (also called cachectin)
  - • TNF suppress appetite, stimulate cell apoptosis
  - ANAEMIA (chronic disease or iron deficiency)
  - HAEMOSTASIS ABNORMALITIES (increased risk for thrombosis)
  - FEVER (due to infection)
  - PARANEOPLASTIC SYNDROME (10%)

CLINICAL EFFECTS OF TUMOURS :

PARANEOPLASTIC SYNDROME

- Symptom complexes in **patient with NEOPLASM** that **cannot be readily explained** either by the local or distant spread of the NEOPLASM.

OR

- By the elaboration of hormone indigenous to the tissue which the NEOPLASM arose.
- It may involve **endocrinopathies**, **nervous system**, **dermatological**, MSK and vascular

CLINICAL EFFECTS OF TUMOURS :

PARANEOPLASTIC SYNDROME

- It may be the early presentation
- May be lethal and mimic metastasis

- **EXAMPLES INCLUDE:**
- Cushing syndrome in carcinoma of the lung (ectopic ACTH syndrome (originally inside PG))
- Polycythaemia in carcinoma of stomach (High [RBC], [Hb], [Hematocrit] in blood)
- Venous blood clots in carcinoma of pancreas
- Finger clubbing in carcinoma of lung

FINGER CLUBBING



- **the tips of the fingers enlarge and the nails curve around the fingertips**, usually over the course of years.
- Nail clubbing is sometimes the result of low oxygen in the blood and could be a sign of various types of lung disease including lung cancer
- When it is related with lung cancer, it is called a paraneoplastic syndrome because its occurrence **cannot be explained** either by the local or distant spread of the NEOPLASM or by hormone production indigenous to the tissue which the NEOPLASM arose

GRADING & STAGING: PROGNOSIS

- **Grading: based on degree of differentiation , architecture and mitosis**
  - This is done Microscopically
  - Categorized as Well- differentiated, moderate & poor
  - It can also be categorized as Low and high-grade
- **Importance of grading:**
  - Well differentiated cancers or Low Grade cancers tend to be less aggressive have a **better prognosis** that the high **grade** cancers
  - Poorly differentiated cancers or High Grade III cancers tend to be more aggressive and have a worse **prognosis** that the lower **grade** cancers

GRADING & STAGING: PROGNOSIS

- **Staging: size, nodal involvement and distant metastasis**
  - Stage refers to the extent of cancer, such as the size of the tumour, and if it has spread.
  - This is the most important prognostic factor
  - Staging is done by clinical examination, radiological and pathological investigation
  - There are many staging systems. Some, such as the TNM staging system, are used for many types of cancer.
  - TNM staging
    - T: tumour size
    - N: lymph node involvement
    - M: extranodal metastasis

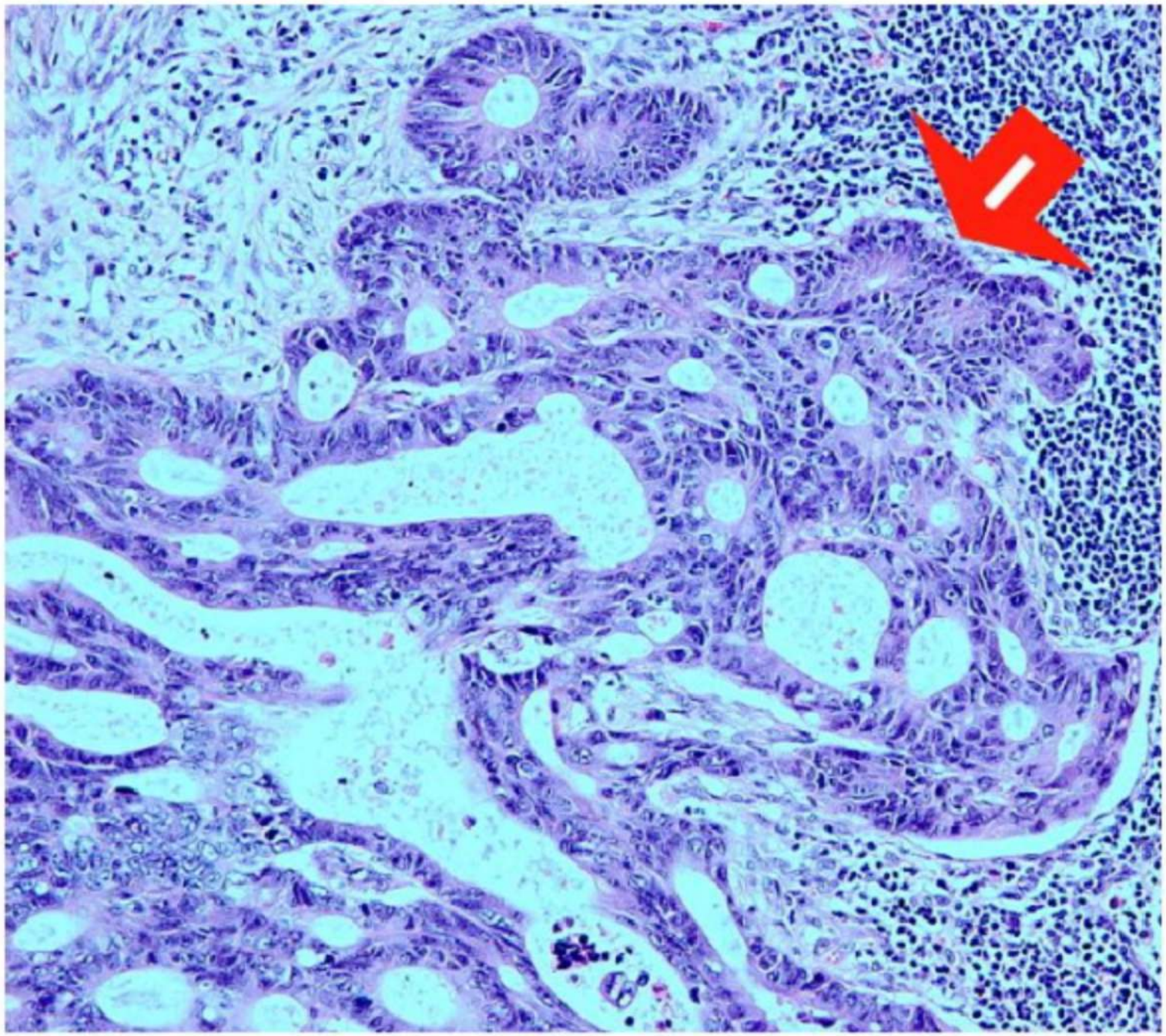
Why Are Cancers Staged?

- The stage suggests the most likely outcome.
  - Knowing the stage gives an educated estimate of life expectancy and the chance of a cure.
- Treatment will be planned and recommended based on the stage of the cancer.
  - For example, surgery is almost always part of the treatment for earlier stage cancers but is not always recommended or possible for advanced or Stage IV cancers.
  - Other therapies, including chemotherapy, radiation therapy, or immunotherapy, are also suggested based on the stage as well as characteristics of the cancer.
- Clinical trials that are available to test and understand new drugs or management strategies are designed to be available only to patients with a specific cancer at a specific stage.
  - It would not be helpful, as part of a trial, to test new drugs on people with Stage I as well as Stage IV cancers. The goals and science are different.

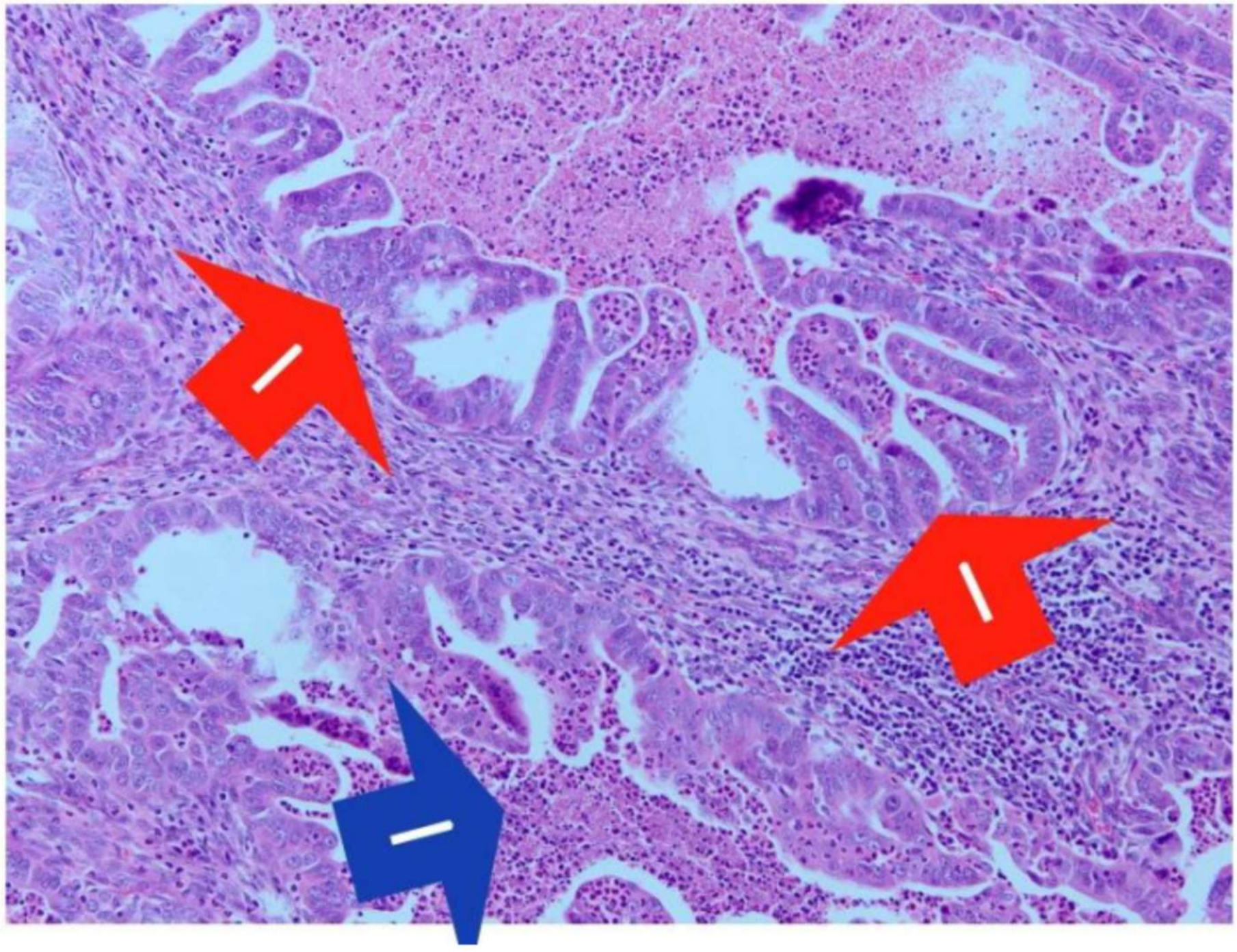
TNM Staging

- **Primary tumor (T)**
  - TX: Main tumor cannot be measured.
  - T0: Main tumor cannot be found.
  - T1, T2, T3, T4: Refers to the size and/or extent of the main tumor. The higher the number after the T, the larger the tumor or the more it has grown into nearby tissues.
- **Regional lymph nodes (N)**
  - NX: Cancer in nearby lymph nodes cannot be measured.
  - N0: There is no cancer in nearby lymph nodes.
  - N1, N2, N3: Refers to the number and location of lymph nodes that contain cancer. The higher the number after the N, the more lymph nodes that contain cancer.
- **Distant metastasis (M)**
  - MX: Metastasis cannot be measured.
  - M0: Cancer has not spread to other parts of the body.
  - M1: Cancer has spread to other parts of the body.

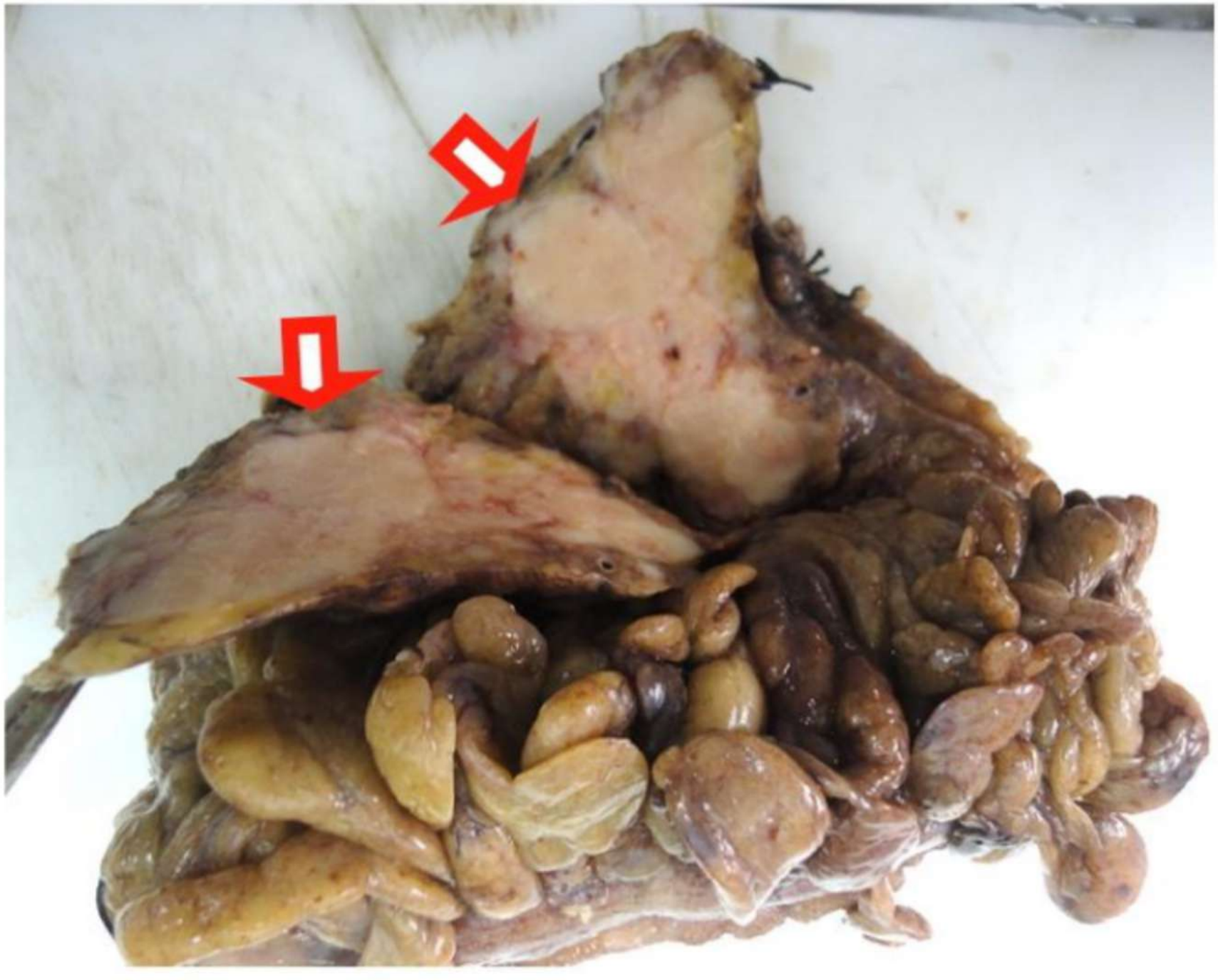
ADENOCARCINOMA AND GLANDS FORMATION



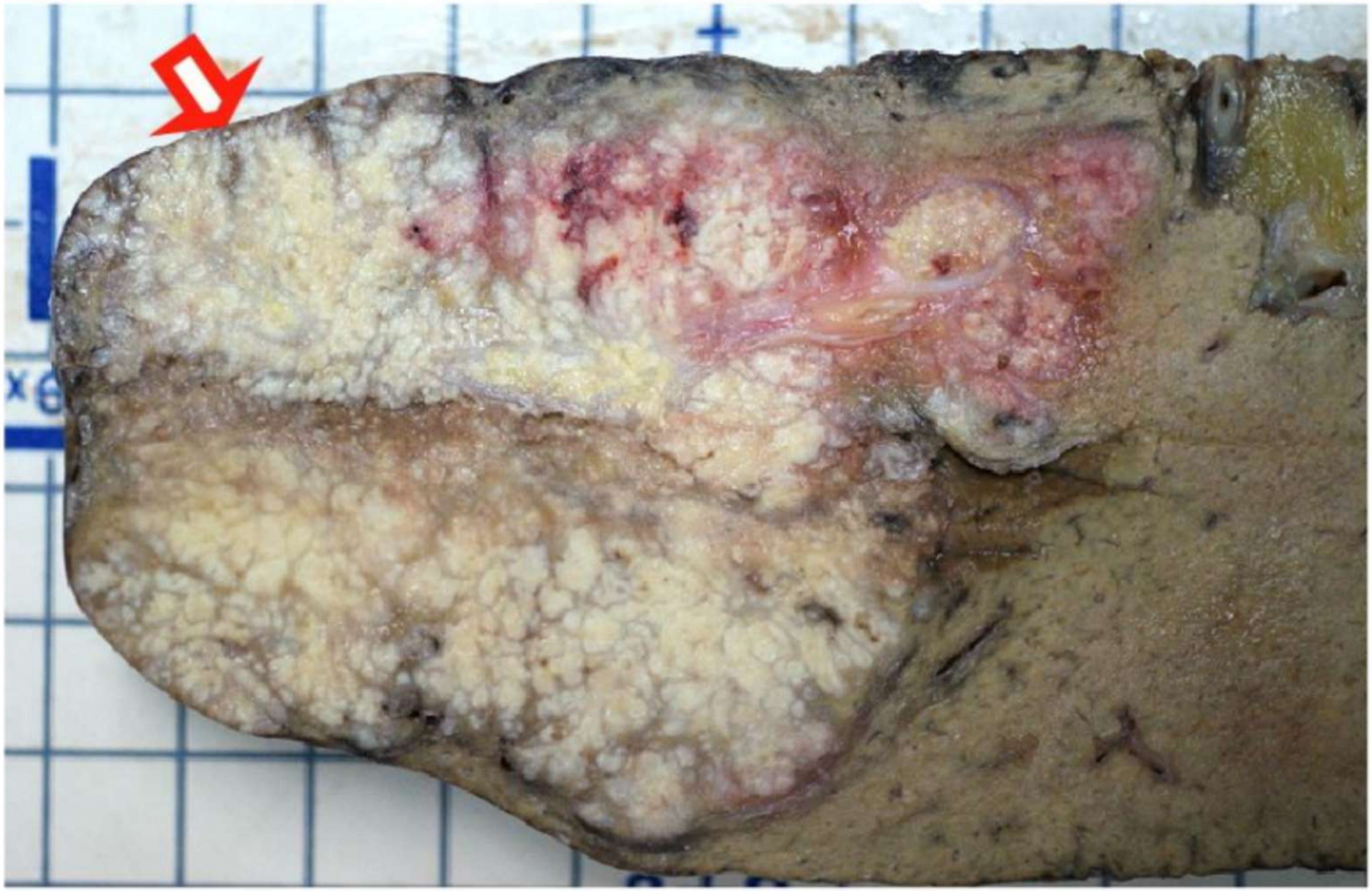
ADENOCARCINOMA AND GLANDS FORMATION



COLONIC CARCINOMA & LYMPH NODE METASTASIS (DUKES C Staging)



COLORECTAL CARCINOMA WITH LIVER METASTASIS (DUKES D Staging)



Part 6: Prevention modalities in cancer

- **Lifestyle modifications**
- Stop smoking cigarettes-the most important factor
  - Cessation of smoking is most important factor in decreasing risk for **cancer of lung, oesophagus, bladder, pancreas, throat, mouth and cervix**
- Increase fiber/decrease dietary saturated animal fat
  - Decreases risk for **colorectal cancer**
- Reduce alcohol intake – decrease the risk of **larynx, oesophagus, colorectal, liver, breast, mouth and throat cancer**
- Reduce weight / keeping a healthy weight
  - (1) Increased adipose tissue increases aromatase conversion of androgens to estrogen.
  - (2) Increased estrogen increases risk for **endometrial and breast cancer.**

Prevention modalities in cancer

- **Immunization**
- **Hepatitis B (HBV) vaccination**
- HBV immunization: ↓ risk for hepatocellular carcinoma
    - Immunization decreases the risk for hepatocellular carcinoma due to hepatitis B-induced postnecrotic cirrhosis.
- **Human papillomavirus (HPV) immunization**
- Human papillomavirus immunization: ↓ risk for cervical cancer
    - Decreases the risk for developing cervical squamous cancer

Prevention modalities in cancer

- **Screening procedures**
- **Cervical Papanicolaou (Pap) smears screening for cervical cancer**
  - Decreases risk for cervical cancer
  - Cervical Pap smear: most responsible for ↓ incidence/mortality rate for cervical cancer

Prevention modalities in cancer

- **Screening procedures**
- **Colonoscopy screening for colorectal cancer**
  - Detects and removes polyps that are precancerous
- **Mammography screening for breast cancer**
  - Detects nonpalpable breast masses
- **Prostate-specific antigen (PSA) screening for prostate cancer**
    - Detects prostate cancer
      - PSA may also be increased in prostate hyperplasia
- **low-dose computed tomography for lung cancer screening**
  - for people who have a history of heavy smoking, and smoke now or have quit within the past 15 years, and are between 55 and 80 years old.

- **Treatment of conditions that predispose to cancer**
- 1. Treatment of *H. pylori* infection:
  - ↓ risk for developing gastric lymphoma/adenocarcinoma
- 2. Treatment of gastroesophageal reflux disease (GERD)
  - Decreases the risk for developing distal adenocarcinoma arising from Barrett's esophagus

TUMOUR MARKERS

- **Biological markers secreted by the neoplastic cells or host response to neoplasm**
- **Used for:**
  - Identification of cancer
    - Some cancer produce specific tumour markers
  - Estimate tumour burden
    - The bigger the tumour, expected higher level of tumour marker produced
  - Detect recurrence
    - After complete removal of the tumour, expected level of tumour marker to be absent / low. If the tumour marker level suddenly increases, recurrence may be suspected
  - Indicator for response to treatment
    - After complete removal of the tumour, expected level of tumour marker to be absent / low.

TUMOUR MARKERS

- Prostatic specific antigen **(PSA)** for prostatic carcinoma
- Human chorionic gonadotrophin **(βHCG)** for choriocarcinoma
- Alpha fetoprotein **(αFP)** in hepatocellular carcinoma
- Carcinoembryonic Antigen **(CEA)** in carcinoma of colon
- Thyroglobulin in thyroid cancer